

Troubleshooting a Business Process

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Background

Introduction

A business process instance is a unique copy of a business process that is created when the business process is run within Sterling B2B Integrator. By monitoring business process instances, you can confirm whether they carry out their intended purposes.

Lesson objectives

This lesson is designed to help you to:

- Check the status of a business process
 - Search a business process
 - View business process details
 - Restart and resume a business process
 - View execution information
 - Terminate a business process
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Viewing Active and Recent Business Processes

Overview

Sterling B2B Integrator provides general and detailed processing information. You can view processing information to monitor active and recent business processes instances to determine whether any further action is required.

Viewing General Processing Information

The Monitor page refreshes automatically and displays the 10 most recent instances of business processes to run and their processing information.

When monitoring active and recent instances, Sterling B2B Integrator shows two status indicators to indicate the required further action:

Status Indicator	Active Business Process	Recent Business Process
	Encountering no errors or warnings at this point of the execution.	Encountered no errors during execution.
	Waiting for other activities to complete before continuing execution. Encountering errors or warnings during execution.	Encountered errors or warnings during execution.

Viewing Detailed Processing Information

The Business Process Detail page provides you with a progress report on a specific business process instance. From the Business Process Detail page, you can also perform activities, such as stopping or restarting an instance.

Exercise 6–1 View General Processing Information

Instructions You can view general processing information by following the steps that are listed:

1. Select **Monitor > Current Processes** from the Sterling B2B Integrator Business Process menu.
2. Use the following fields and columns to view general processing information about instances and perform other activities in the Monitor page, as appropriate:

Field/Column	Description
Automatically refresh every minute	Default time to refresh the list of the 10 most recent instances. To disable this feature, clear the check box.
Status	Indicator of the status of an active or recently run instance.
ID	Number that is assigned by Sterling B2B Integrator to identify an instance. Click the number to display the Business Process Detail page.
Name	Name of an instance. Click the name to view the BPML code that makes up the instance.
State	Current state of an instance. The following list shows possible states in the order of precedence during branch processing: ■ Active/Running ■ Completed ■ Terminated ■ Waiting ■ Interrupted ■ Halting/Halted
Continued on next page	

Exercise 6–1 View General Processing Information (Continued)

**Instructions
(Continued)**

Field/Column	Description
Started	Date and time an instance started.
Ended	Date and time an instance ended.
Deadline	Deadline by which the instance must run, as specified when setting up a schedule for the business process.
Expires	Information about when an instance expires. Click Info to display the expiration information, including whether the data for the instance is archived after it expires.
Parent/Child	Parent or child business process that is referenced when running an instance. Click the up arrow to view a parent business process. Click the down arrow to view a child business process.

Exercise 6–2 Viewing Detailed Processing Information

Instructions You can view detailed processing information by following the steps that are listed:

1. Select **Monitor > Current Processes** from the Sterling B2B Integrator Business Process menu.
2. Click the number that identifies the instance in the ID column in the Monitor page.
3. Use the following fields to review detailed processing information and perform activities in the Business Process Detail page, as appropriate:

Field/Column	Description
Name	Name of the instance for which you are viewing details. Click the name to view the BPML code that makes up the instance.
Instance ID	Number that is assigned by Sterling B2B Integrator to identify this instance.
Status	Current status of this instance. Possible status levels are: ■ Success ■ Error
Continued on next page	

Field/Column	Description
State	Current state of this instance. The following list shows possible states in the order of precedence during branch processing: ■ Active/Running ■ Completed ■ Terminated ■ Waiting ■ Interrupted ■ Halting/Halted
User	Identification of the user who ran the process.
Continued on next page	

Exercise 6–2 Viewing Detailed Processing Information (Continued)

Instructions (Continued)

Field/Column	Description
Action	List of actions to complete for this instance, including an activity to generate an XML report. The actions available in this field are determined by whether the instance is currently active or stopped. Possible activities are: ■ Restart – Continues running an instance. ■ Stop – Stops running an instance. ■ Terminate – Cancels an instance and all remaining active and waiting subprocesses. ■ XML report – Generates an XML report that describes the instance. If you terminate an active instance, the State field indicates with messages in the following order: Halting > Halted > Terminated.
Step	Current step of this instance.
Service	Name of the service that is running for a current step. Click the service name to view settings for a service in this instance.
Status	Current status of the steps in this instance. Possible status levels are: ■ Success ■ Error
Advanced Status	Service-specific details about any errors that occurred for a step in this instance, when applicable. Click the message to display information.
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Exercise 6–2 Viewing Detailed Processing Information (Continued)

**Instructions
(Continued)**

Field/Column	Description
Started	Date and time the step of the instance started.
Ended	Date and time the step of the instance ended.
Status Report	Status report that provides the results of the step. To view the status report, click info.
Document	Business process document that this service is processing (that is, the primary document). To view the document, click info.
Instance Data	Contents of the process data that are generated after a specific step in this instance. In addition, this field links to any messages that are going to or coming from a service. To view the information, click info.

Searching for Business Processes and Other Information

Overview

In Sterling B2B Integrator, you can use the Central Search page to search for information about:

- Additional live (active) business process instances that do not display in the Monitor page
- EDI Correlation matches

In the Central Search page, you can specify the search criteria to locate business process instances by business process name, status, or by date and time.

Searching for Business Process Instances (Advanced)

You can perform an advanced search for the following information:

- BPSS business flow transactions
- Active, archived, and restored business processes
- Document Tracking
- EDI records for business process instances that included EDI interchange processing
- Correlation name-value pairs that you defined using the Correlation service in a business process or services, such as EDI and CII services, where Sterling B2B Integrator generates correlations
- EDIINT transactions

Sterling B2B Integrator also allows you to search for business process instances by:

- Location of business process
 - Business process ID
 - Business process name
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Exercise 6–3 Searching for Business Process Instances (Central Search)

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- | | |
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| Instructions | <p>You can perform a basic search for business process instances by following the steps that are listed:</p> <ol style="list-style-type: none">1. Select Monitor > Central Search from the Sterling B2B Integrator Business Process menu.2. Enter the yesterday date in the form MM/DD/YYYY in the Start Date From dialog box, and time 01:00:00 (HH:MM:SS).3. Click Go!4. Click the number link that indicates the number of matches in Sterling B2B Integrator in the Central Search Results page. <p>The Monitor page opens, listing the business process instances that match your search criteria.</p> |
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Exercise 6–4 Searching for Business Process Instances (Advanced)

Instructions

You can follow the steps that are listed to perform an advanced search for business process instances:

1. Select **Monitor > Advanced Search > Business Processes** from the Sterling B2B Integrator Business Process menu.
2. Specify any combination of the following search criteria in the Business Process Monitor Advanced Search page, as appropriate:

Field	Description	Action
Search Location		
Select the area to search from	Instances that are maintained in a specific location.	Select one of the following options: Live Tables – Display live (active) instances. Restored Tables – Display data for instances that is restored from an offline location.
Search Using Business Process ID		
Process ID	ID assigned by Sterling B2B Integrator to identify an instance.	Type the ID for an instance.
Search Using Business Process Name		
Business Processes	List of instances that are currently maintained in Sterling B2B Integrator.	Select an instance from the list.
Continued on next page		

Exercise 6–4 Searching for Business Process Instances (Advanced) (Continued)

**Instructions
(Continued)**

Field	Description	Action
System Business Processes	Sterling B2B Integrator system business processes (that is, business processes that complete or have completed system operations).	Select a system business process from the list.
State	Current or final state of an instance.	The default value is ALL (displays all instances). Maintain the default value or select one of the following options: ■ Completed ■ Waiting ■ Active ■ Halted ■ Halting ■ Interrupted ■ Terminated
Continued on next page		

Exercise 6–4 Searching for Business Process Instances (Advanced) (Continued)

**Instructions
(Continued)**

Field	Description	Action
Status	Current or final status of an instance.	The default value is ALL (displays all instances). Maintain the default value or select one of the following options: ■ Success ■ Error
Start date/time range	Instances running or completed within the specified start dates and times.	Type a starting date and time range and select A.M. or P.M.

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Exercise 6–4 Searching for Business Process Instances (Advanced) (Continued)

**Instructions
(Continued)**

3. Click **Go!** The Monitor page opens, listing the business process instances that match your search criteria.

The following table lists the other available Advanced Search options:

Advanced Search Option	Search by:
BPSS Correlation	Location, Transaction Type, Trading Partner, Status, or Date/Time
Correlation	Type, Location, and Name/Value pairs
Documents (Document Activity search)	Location, Sender, Receiver, Type, Status, Tracking ID, Correlations, or Document ID
EDI Correlation	Location, Test Mode, Interchange, Group, or Transaction level specifics
EDIINT	Contracts, Status, Type, or dates

Searching for Current Service Activities

Introduction

This section allows you to search for service activity by service type, service name, activity type, Not Updated Since date, or Started Before date.

Follow the steps that are listed to perform a search of service activity:

1. Select **Monitor > Current Activities** from the Sterling B2B Integrator Business Process menu.
 2. Enter yesterday date in the form MM/DD/YYYY in the Not Updated Since dialog box, and time **01:00** (HH:MM).
 3. Click **Go!**
 4. View the list of the business process instances that match your search criteria in the Service Activity Search Results page.
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Restarting or Resuming a Business Process

Introduction

The Restart Business Process page opens after you select Resume or Restart in the Action field from the Business Process Detail page. This page has three options that allow you to determine how to run a business process again if it is stopped running: Resume, Simple Restart, and Advanced Restart.

Important!	Resume appears only if the instance stopped in mid-execution.
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The Troubleshooter page allows you to start running multiple business processes at the same time.

Resume

The Resume function restarts a business process from the last error step. The business process retains its original ID. Use this function when the error is related to services. For example, if a service is disabled unexpectedly, a business process using that service stops at that point. After the service is restored and active, you can resume the business process.

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Restarting or Resuming a Business Process (Continued)

Simple Restart

The Simple Restart function starts a new instance of a business process from the beginning. The new business process receives a new ID. Use this function when the error is related to an unplanned interruption of the business process. For example, if you have interrupted a business process while bringing down a server for non-routine maintenance, you can restart the business process after the server is restored by using the Simple Restart function. This ensures a complete execution of the transaction.

Advanced Restart

The Advanced Restart function runs the original document using a different version of the business process or another business process. The new instance receives a new ID. Use this function when the error is related to the business process BPML code. For example, if a business process has incorrectly routed a document through sequential services, you can modify the business process and enable a new version to correct the error. You then use the new version of the business process to run the original document.

Exercise 6–5 Resuming a Business Process or Performing a Simple Restart

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|---------------------|--|
| Instructions | <p>You can follow the steps that are listed to resume a business process or to perform a simple restart:</p> <ol style="list-style-type: none">1. Select Monitor > Current Processes from the Sterling B2B Integrator Business Process menu.2. Click the number that identifies the instance in the ID column in the Monitor page.3. Select Restart in the Activities field in the Business Process Detail page, then click Go!
The Restart Business Process screen opens.4. Click Go! for the Simple Restart.<ul style="list-style-type: none">■ To resume the business process from the point at which it stopped, in the Resume area, click Go! The instance retains its original business process instance ID.■ To restart the business process from the beginning, in the Simple Restart area, click Go! The new business process receives a new instance ID. <p>A new page opens for the selected activity. Ensure that the business process runs to completion. If the resumed or restarted business process fails, you must find and correct those errors before the business process can run to completion.</p> |
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Exercise 6–6 Performing an Advanced Restart

Instructions

You can follow the steps that are listed to perform an advanced restart.

1. Select **Monitor > Current Processes** from the Sterling B2B Integrator Business Process menu.
2. Click the number that identifies the instance in the **ID column** of the Monitor page.
3. Select **Restart** in the Action field list in the Business Process Detail page, and click **Go!**
4. Select the business process that you want to use for this instance in the Advanced Restart area.
5. Click **Go!** in the Advanced restart area.
The business process restarts in a new window, using the selected business process. The new business process receives a new instance ID.
6. View any system information or reports that are provided as the business process runs.

Ensure that the business process runs to completion. If the restarted business process fails, use the reports and system information to identify the problem.

Viewing Execution Information

Overview

When you restart or resume a business process, Sterling B2B Integrator opens the Execute Business Process page. This page lists the service, status, advanced status, start and end times, troubleshooting reports, and documents for each step that are completed within a recently run business process.

Important!

The Information icon  indicates that a troubleshooting report or document is available.

Terminating Business Processes

If a business process stops running, you can also use the System Troubleshooter page to review the business process and terminate it when you see a status of waiting, interrupted, or halted. The System Troubleshooter page allows you to terminate processing for multiple business processes. Note the difference between terminated business process and halted business process.

Halted	Terminated
The Business process stops because of an error occurred during execution and no on-fault is provided.	The business process stops because of manual intervention.
The business process can be resumed from the step where it stopped.	The business process can be restarted and cannot be resumed from the step where it stopped.

Walk-through: Terminating a Business Process

Task	You are required to review the business process and terminate, if necessary.
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Instructions	You can follow the mentioned procedure to review the business process and terminate, if necessary. 1. Select System > Troubleshooter from the Operations menu. 2. Check the status for Business Process Usage in the System Troubleshooter page, under System Status. When running business processes show any of the following statuses, the status becomes an active link: ■ Active/Running ■ Interrupted ■ Interrupted_Man - Manually ■ Interrupted_Auto - Automatically ■ Halted 3. Click the status link. For example, if Business Process Usage displays: Interrupted: 9 (where 9 is the number of interrupted business processes) 4. Click Interrupted to display the Interrupted Business Processes dialog box. 5. Click the check box under Select in the dialog box; next to the business processes to terminate. 6. Select Terminate from Activities and click Go! 7. Click OK when Sterling B2B Integrator prompts you to verify that you are sure you want to terminate the business processes. 8. Review the report to verify that Sterling B2B Integrator terminated the business processes.
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Important!	You cannot terminate business process with a status of Halting.
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Exercise 6–7 Terminating a Halted Business Process via Operations, System, Troubleshooter

Instructions

In this exercise, we will run the BasicInventory business process without an input file which will create a Halted business process and terminate that process via the Operations > System > Troubleshooter screen.

1. Type **BasicInventoryProcess** in the Search text box from the Business Process Manager screen and click **Go!**
2. Click the **Execution Manager** icon for BasicInventoryProcess.
3. Click the **execute** icon.
4. DO NOT specify an input file name and click **Go!**
5. Observe the separate Execute Business Process window while the business process runs.
6. Close the **Execute Business Process** window when the Business Process has completed running.
7. Click the **Business Process Usage** link from Operations > System > Troubleshooter.
8. Click the link that is to the right of the Halted status.
9. Select or check the box next to BasicInventoryProcess.
10. Select **Terminate** from the Activities drop-down list and click **Go!**
11. Click **OK** to confirm that you want to terminate the business process.
The Sterling B2B Integrator system returns a message, “Request to terminate the business process has been successfully submitted.”
12. Close the **Terminating Business Processes Report** and the Business Process Usage windows.

Exercise 6–8 Terminating a Halted Business Process via Business Process Monitor, Current Processes

Instructions

In this exercise, we will run the BasicInventory business process without an input file which will create a Halted business process and terminate that process via the Business Process Monitor > Current Processes screen.

1. Type **BasicInventoryProcess** in the Search text box from the Business Process Manager screen and click **Go!**
2. Click the **Execution Manager** icon for BasicInventoryProcess.
3. Click the **execute** icon.
4. DO NOT specify an input file name and click **Go!**
Observe the separate Execute Business Process window while the business process runs.
5. Close the **Execute Business Process** window when the Business Process has completed running.
6. Locate the Halted **BasicInventoryProcess.bp** from Business Process Monitor > Current Processes and click the business process ID link.
7. Select **Terminate** from the Action drop-down list from the Business Process Detail screen, and click **Go!**
8. Click **OK** to confirm that you want to terminate the business process.
The Sterling B2B Integrator system returns a message, “Request to terminate the business process has been successfully submitted.”
9. Click **Return** to return to the Business Process Monitor page. Notice that the state of BasicInventoryProcess is now Terminated.

Quiz

Questions

1. In current processes, if the status shows a green traffic light, the state will always be completed successfully.
 - a. True
 - b. False
 2. Which field in the Execute Business Process page has the process data that are generated in each step?
 3. When does the Parent/Child column of the Process Monitor indicate "none"?
 - a. This process does not invoke any process and no other process invoked this process.
 - b. This process failed to read in a designated file.
 - c. This process failed to write output to the designated directory.
 - d. This process failed has not received input file to invoke the other process.
 4. The Terminated business process can be resumed from the step where it stopped.
 - a. True
 - b. False
 5. At what state should a business process be, for it to be archived? Select all that apply.
 - a. Completed
 - b. Terminated
 - c. Halted
 - d. Interrupted
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Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Check the status of a business process
 - Search a business process
 - View business process details
 - Restart and resume a business process
 - View execution information
 - Terminate a business process
-

Introduction to Service Management

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Background

Introduction	This lesson provides an overview of service management parameters.
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Lesson objectives	This lesson is designed to help you to: ■ List and explain the four basic service types ■ Explain how to create a service configuration ■ Explain Bootstrapping ■ Describe how the file system adapter works ■ Describe how OnFault Processing works ■ Explain how the Command Line adapter functions ■ Describe and demonstrate the use of a Lightweight JDBC ■ Describe how the Invoke Business Process service works
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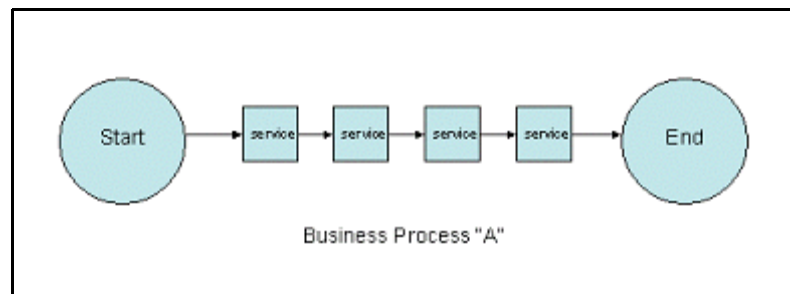
Difference Between Services and Adapters

Introduction to Services

What makes Sterling B2B Integrator so flexible is the capabilities of its services. A service is a small program that B2Bi can call to accomplish tasks in a bp. Usually, the high-level purpose of a service is to achieve some pre-defined type of integration activity, such as:

- Communicating with external applications or middleware (by way of special services called adapters)
- Performing data manipulations, such as translation, transformation, splitting, and joining
- Routing data based on payload, content, or meta data
- Publishing data to interested subscribers, which can trigger a new business process instance or allow an executing process to continue
- Executing one or more B2B protocols
- Spawning a pre-defined business process (nested process)

Services that communicate with external systems are called adapters. Input Services and Output Services are adapters. A service that runs inside Sterling B2B Integrator and does not touch outside systems will retain the service title in its name.



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Difference Between Services and Adapters (Continued)

Introduction to Services (Continued)

Consider that a business process involves determining how many open orders a specific customer might have. The activities might include:

- Looking up the customer number
- Retrieving all orders for that number from an orders database
- Removing all but pending orders from that table
- Reporting the information in printed form

Each of these activities could be accomplished through a service.

Sterling B2B Integrator includes a wide variety of services, of which some are internal, while others are external and configurable. It provides a base which is a set of services, which can be deployed and then extended as your needs evolve and new technologies emerge. Custom services can also be developed to support your unique needs.

Adapters (Reach Outside Sterling B2B Integrator)

Adapters can appear at any point of a business process and are often used to initiate business processes. For example, a File System Adapter could poll a directory to detect any new files (purchase orders) and initiate a business process that processes the input. If the adapter is set up to perform only one particular task, its usage is limited. When we look at setting up adapters for use, we will create instances of adapters to define their settings.

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Difference Between Services and Adapters (Continued)

Service Types

These are the basic service types in Sterling B2B Integrator:

- Internal Services that are totally inside Sterling B2B Integrator. While internal services accept parameters and produce results, they do not directly interact with outside systems. These types of services will usually retain “Service” in their name. For example the Translation Service.
- Input Services that receive data from outside systems.
- Output Services that send data to outside systems.

Input and output services are collectively known as adapters. Adapters interact with the world outside of the Sterling B2B Integrator J2EE environment.

Adapters can appear in the middle of business process definitions or at the end. For a complex application adapter, for example the SOAP adapter, calls to the adapter are actually calls to an underlying Sterling B2B Integrator-defined SOAP business process. As you can see, services themselves can be implemented as business processes.

Service Configurations

The services you include in your business process model may require configuration in order for your process to run correctly. You configure some parameters in Sterling B2B Integrator interface (this is called creating a service configuration), some in either the interface or the GPM, and some exclusively in the GPM. A few require manual editing in the BPML code. The reference documentation for each service describes the parameters for the service and indicates where to perform the parameter configuration.

Generally, you can configure service parameters either during initial service configuration in Sterling B2B Integrator interface, or during business process creation using the GPM. You can modify service configuration parameters at any time.

File System Adapter

Overview

The File System adapter (FSA) is functionally the simplest of all the Sterling B2B Integrator adapters. It collects files from a disk and can be used to initiate business processes. It also extracts files from Sterling B2B Integrator back to a disk. It can retrieve and send any type of file, but it does not actually read the file.

How the File System Adapter Works

Use the File System adapter to collect (import) files from a file system into a business process and extract (export) files from a business process to a file system. You can configure the File System adapter to start a business process after files are collected from the file system or include the File System adapter in a business process flow. In addition, you can schedule the File System adapter to run at specific time intervals.

You can set a file system adapter configuration to collect, extract, or both. However, each instance of a File System adapter in a business process will either **collect or extract**, it can not do both at the same time. Normally, you should use only the extraction side of the File System adapter in a business process as the collection is usually done by a bootstrapped configuration designed to kickoff a bp rather than being part of one.

The FSA works with the file system of the server where the configuration is active. In a clustered environment the adapter can only run on a specific node, unless there is a type of shared file storage. Other file systems can be used but they must be mounted as part of the local system

Collection

When a FSA is set to Collection there are two types of modes, each one causing different behavior:

- Bootstrapped - Is designed to have the FSA monitor a folder based on a schedule and start a business process when a file is collected.
- Non-Bootstrapped - Is designed to be part of a business process and collect a file as part of the bp flow.

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File System Adapter

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How the File System Adapter Works (Continued)

Extraction

The activity engine calls the extraction portion of the file system adapter when the file system adapter is placed in a business process. When called, the adapter handles extraction requests in a parallel, asynchronous manner.

When the adapter is called in a workflow context, it:

1. Creates a thread.
2. Checks to see whether the instance is enabled for extraction.
 - If disabled, it creates an audit entry and nothing further is done.
 - If enabled, it:
 - Queries the service controller for the properties that are defined in the activity information portion of the workflow.
 - Extracts the data to the specified folder. The file that is created is given with the specified name.
 - Tries to shut down the RMI server.

When a FSA is used to extract a file to disk in a bp it will extract Primary Document. The adapter can be configured to use the original name of the file picked up prior or have the adapter create a new one based on a setting or variables.

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File System Adapter

(Continued)

Configuring a File System Adapter

After you install the adapter, you need to create an instance configuration in Sterling B2B Integrator before you can use the adapter.

1. You can configure the file system adapter in the following manner.
2. Select **Deployment > Services > Configuration** from the Sterling B2B Integrator interface.
3. Locate the **Create** section on the Services Configuration screen. Click **Go!** for New Service.
4. Click the **List view** icon on the Select Service Type screen.
5. Scroll through the list and select **File System Adapter** as the type of service you want to configure and click **Save**.
6. Click **Next** to proceed.
7. Provide a unique and meaningful name to the service configuration.
8. Provide a meaningful description for reference value.
9. Select **None** in the Select a group area.
10. Click **Next** to proceed.
11. Select the folder or subfolder on the SAME computer as the installation of Sterling B2B Integrator for collection, as this is safer. However, the folders or subfolders can exist on a different computer as long as the Sterling B2B Integrator computer can access it, including all necessary permissions.



Caution

Do not specify a folder that contains the programs or any files you do not want to lose. The file system adapter does not copy the files that it collects for processing.

If you wish to collect only files with a specific file extension within the specified folder, use the Filename filter field. You can also use this field to specify a single file to be collected (that is, File1.txt). Additionally, you can use the Filename filter field with wildcards to collect a subset of files within a folder (that is, *.txt).

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File System Adapter

(Continued)

Configuring a File System Adapter (Continued)

- a. Do you want to collect only from the folder or also from subfolders?
 - Yes - collects files from the folder and all subfolders
 - No - collects files only from the folder specified
 - b. Do you want to use absolute filename for the document name?
 - Yes - keep the absolute file path name of the files collected when assigning the document name
 - No - default
 - c. Do you want this adapter to start a business process?
 - Yes - the business process will be specified on the next screen
 - No - no business process will be started
 - d. Click **Next** to proceed.
12. If you specified that the file system adapter collection should begin a business process, you must select the business process for it to initiate.
- a. Select the correct business process.
 - b. Select the type of document storage you want to use.
 - System Default
 - Database
 - File System
13. Specify whether to obscure the file contents when collecting. Does not work with "attachFile" or "importFile".
- Yes - File contents will be obscured
 - No - File contents will not be obscured

(Continued on next page)

File System Adapter

(Continued)

Configuring the File System Adapter (Continued)

Important!	This field displays only as an option if Start a business process once files are collected is set to Yes. Additionally, if you specify this parameter using the configuration, you cannot override this value using the Graphical Process Modeler Obscure option. However, you can override this parameter using BPML. Will the file contain Obscure File Contents?
-------------------	---

- a. Specify a User Parameter 1: A user parameter that is passed to the bootstrapped workflow and placed in process data as UserParm1.
- b. Specify a User Parameter 2: A user parameter that is passed to the bootstrapped workflow and placed in process data as UserParm2
- c. Specify a User Parameter 3: A user parameter that is passed to the bootstrapped workflow and placed in process data as UserParm3
- d. Specify a User Parameter 4: A user parameter that is passed to the bootstrapped workflow and placed in process data as UserParm4
- e. Specify a User Parameter 5: A user parameter that is passed to the bootstrapped workflow and placed in process data as UserParm5

Important!	These fields display only as an option if Start a business process once files are collected is set to Yes. Additionally, if you specify this parameter using the configuration, you cannot override this value using the Graphical Process Modeler userParmX option. However, you can override this parameter using BPML.
-------------------	---

- f. Click **Next** to proceed.
-

File System Adapter

(Continued)

Configuring a File System Adapter (Continued)

14. This screen allows you to select who runs it and how it is invoked.
 - a. Choose a login ID for the Run As User: dialog box.
 - b. Use 24 Hour Clock Display? - If selected, the adapter will use the 24-hour clock instead of the default 12-hour clock.
 - c. Under Schedule: you can choose:
 - Do not use schedule - This is for adapter instances that will be used in business processes
 - Run based on timer - Schedule the adapter to collect data based on a specific time period and at startup.
 - Run daily - Schedule the adapter to run at the same time every day
 - Run based on day(s) of the week - Schedule the adapter to run on specific day (s) of the week. You can specify any date exclusions.
 - Run based on day(s) of the month - Schedule the adapter to run on a specified day(s) of the month at a certain time.
 - d. Click **Next**.

(Continued on next page)

File System Adapter

(Continued)

Configuring a File System Adapter (Continued)

15. Select the folder for the file system adapter to write files to as part of a business process for extraction.
 - a. Assign a File name to be given for the extracted files.
 - b. Specify whether to Unobscure the file contents when extracting. Does not work with "exportFile".
 - Yes - File contents will be unobscured
 - No - File contents will not be unobscured

Important!	This parameter is read-only in the Graphical Process Modeler.
-------------------	---

- c. Select a file naming convention:
 - Use the original file name as the extracted file name - keeps the names of the files. Go to step 12.
 - Assign a specific name - Give the file the name specified on the following screen in step 11.
 - d. Click **Next** to proceed.
16. Confirm that all the information entered is correct on the Confirm screen. If needed, use **Back** to return to previous screens to change information. Once all the entries are correct, click **Finish**.

(Continued on next page)

File System Adapter

(Continued)

Configuring File System Adapter in the Graphical Process Modeler

When the file system adapter is placed into a business process, you must select a configuration instance. The following table describes the fields that are used to configure the File System adapter in the Graphical Process Modeler. This table contains only the fields that are configured in the Graphical Process Modeler. The values in parentheses represent the corresponding BPML values.

Field	Description
Config (participant name)	Name of the adapter configuration. Required. No default.
Action	Action that the File System adapter is to perform. Required. No default. Valid values: ■ Collection (FS_COLLECT) - Files are collected or picked up from the specified folder. ■ Extraction (FS_EXTRACT) - Files are extracted or written to the specified folder.
appendOnExtract	Whether to append the data if the extract file exists. Normally, files are overwritten when extracting. This parameter allows you to append the data to the existing files instead. Valid values: ■ Yes (true) - Data is appended to existing files. ■ No (false) - Existing files are overwritten. Default
attachFile	Used to attach a file to a workflow as the primary document. The adapter does not perform any I/O and does not delete the file. Any valid filename is a valid value.
Continued on next page	

File System Adapter

(Continued)**Configuring File
System Adapter in
the Graphical
Process Modeler
(Continued)**

Field	Description
checkDelete	Determines whether checking for deletion is possible before collecting files. Valid values: ■ Yes (true) - Default ■ No (false)
collectMultiple	Used to collect multiple files in non-bootstrap mode. Collected files are placed into process data. ■ Yes (true) ■ No (false) - Default
collectMultiplePDname	Used when collecting multiple files to specify which file will be the primary document. Any valid filename is a valid value.
	Continued on next page

File System Adapter

(Continued)**Configuring File
System Adapter in
the Graphical
Process Modeler
(Continued)**

collectMultiplePrefix	Specifies a prefix to be added to the document name. When multiple documents are created in process data, the documents are named Document1 through DocumentX. Multiple instances could overwrite the documents. You can use this prefix to differentiate the documents in different instances. For example, One instance could use the prefix Inst1_ and another instance could use the prefix Inst2_. The first instance would produce files named Inst1_DocumentX and the second instance would produce files named Inst2_DocumentX. The actual file name is placed as an attribute (filename=) in the document tag. The default value is FSA_.
collectZeroByteFiles (true/false)	Whether to collect zero-byte files. Valid values: ■ Yes (true) - Zero-byte files are collected. ■ No (false) - Zero-byte files are ignored. Default.
Continued on next page	

File System Adapter

(Continued)

Configuring File System Adapter in the Graphical Process Modeler (Continued)

Field	Description
concatenateFiles	Used when the collectMultiple option is true and when the File System adapter is set in a non-bootstrap mode. The content of multiple non-zero byte files is concatenated into a single file, and is placed as the primary document. Optional. Valid values: ■ Yes (true). Default. ■ No (false)
dbCollect	If you set this field to true and the field deleteAfterCollect is set to true (which is the default), a database record will be written for every file collected. Before a file is collected, the database is checked to see whether the file has already been collected. Optional. Valid values: ■ Yes (true) ■ No (false). Default
dbPurgeCollectMi	Used when dbCollect is set to Yes (true) to specify the number of minutes, from the time the database record is written, before the record is purged. Set this value slightly higher than the scheduled collection interval to prevent duplication before purging. Optional. Valid value is any valid (positive) integer value. Default is 1440 (one day).
Continued on next page	

File System Adapter

(Continued)

Configuring File System Adapter in the Graphical Process Modeler (Continued)

Field	Description
deleteAfterCollect	Whether to delete the file after collection. Valid values: ■ Yes (true) - File is deleted from the Collection folder after it is collected. Default. ■ No (false) - File is left in the folder after it is read into Application
fileModTimeThreshold	Sets the file modification time threshold (in seconds) for files to be collected. A file is collected only if the modification time of the file is older than the number of seconds specified. This prevents premature collection of a file. Defaults to 30 seconds if you do not specify a value.
Continued on next page	

File System Adapter

(Continued)

Configuring File System Adapter in the Graphical Process Modeler (Continued)

Field	Description
filter	Collect only files that match a specified filter within the collection folder. Optional. Examples include: ■ *.txt (collects only .txt files) ■ *.dat (collects only .dat files) ■ EDI.* (collects only files named EDI with any file extension) ■ EDI.txt (collect only files named EDI with a file extension of .txt) Note: If there are multiple files in the collection folder and you leave this field blank, one of the following occurs: If the adapter is configured to start a business process, it processes all files placed in the collection folder. If the adapter is within a business process, it collects only the first file in the collection folder. Note: If you specified this option using the File System adapter configuration, this field will be read-only.
genReport	Determines whether a workflow status report is generated for all files regardless of whether they were successfully collected or not. Optional. Valid values: ■ Yes (true) - Status report will be generated whether file collection was successful or unsuccessful. Default ■ No (false) - Status report will only be generated if file collection is unsuccessful
maxCollect	Sets the maximum number of files to collect. The default is -1 (unlimited)
Continued on next page	

File System Adapter

(Continued)

Configuring File System Adapter in the Graphical Process Modeler (Continued)

Field	Description
maxThreads	Used for performance tuning to set the maximum number of threads that are used when collecting files. The default is 10 threads.
noFilesSetSuccesses	Used to determine the workflow status when no files are available to collect in non-bootstrap mode. Optional. Valid values: ■ Yes (true) - The workflow status is set to Success even if no files exist in the specified collection folder during collection. ■ No (false) - The workflow status is set to Error if no files exist in the specified collection folder during collection. The default is No (false)
sortBy	When collectMultiple option is true and the File System Adapter is configured in a non-bootstrap mode, the files are sorted by File name or Modified date as selected in the Graphical Process Modeler. The default setting places the files the way they were placed in the Collection folder. Optional. Valid Values: none.
streamBufSize	Used for performance tuning to override the default buffer size of 5k (5120). Optional. Valid value is any integer.
subCharsOnExtract	If the document name contains invalid filename characters, you can use this field to have them replaced with something else. An example would be if the document name was a GUID that contains colons `:` which are invalid in a Windows filename. In this case, you would enter ":_ " to replace all occurrences of the colon with an underscore. Optional. Must be entered in two character pairs without delimiters or spaces. The first character is the one to be replaced, the second is the replacement itself.

Instructor Note: Inform students with the Linux login details and, help the students in using linux box. Share the location details of all the data files that are used in the exercises.

Exercise 7–1 File System Adapter

Instructions

For ease of understanding, this procedure is chunked into three parts that are as listed:

- Create the Service Configuration
- Create a Business Process
- Check in and Test the Business Process

Important!

Example: Create two new directories in the Linux home directory(/home/train1):

/home/student/fscoll

/home/student/fsext

Create Folders

You need to create extraction and collection folders on the Linux server.

1. Open Putty and login to SI instance using root account
2. Go to the student directory with the following command:
`cd /home/student`
3. Create two new folders using the following commands:

```
mkdir fscoll fsext
```

(Continued on next page)

Exercise 7–1 File System Adapter

(Continued)

Create the Service Configuration

You can create the service configuration in the following manner.

1. Log in to admin console.
2. Click **Deployment > Services > Configuration** on the Admin console.
3. Click **Go!** in the new service box.
4. Type **File System Adapter** in the Service Type: dialog box and click **Next**.
5. Use **FS_Test** for a name and Test file system adapter for a description. Do not create or select a group. Click **Next**.
6. Specify **/home/student/fscoll** for the collection folder.
 - a. Leave the filename filter blank.
 - b. Collect files from sub-folders within and including the collection folder: **No**
 - c. Use the absolute file path name for the document name: **No**
 - d. Start a business process once files are collected: **No**
 - e. Click **Next**.
7. Specify **/home/student/fsext** for the extraction folder.
 - a. Unobscure File Contents?: **No**
 - b. File naming convention: Assign a specific name
 - c. Click **Next**.
8. Enter **%^output.txt** for the output filename and click **Next**.
9. Ensure that the service is enabled for business processes and click **Finish**.

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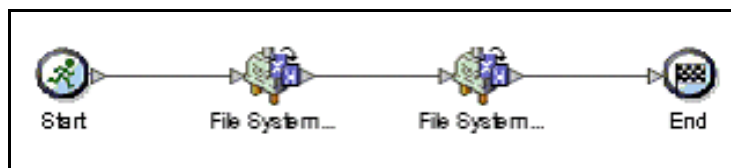
Exercise 7–1 File System Adapter

(Continued)

Create a Business Process

You can create a business process by following the steps that are listed:

1. Open the Graphical Process Modeler. If it is already open, click **View > Refresh Services**.
2. Click **File > New**.
3. Drag a **Start**, **End**, and two **File System Adapter** stencils into the work area, and connect them as shown in the following figure.



4. Open the Service editor for the first file system adapter and choose **FS_Test** for the configuration.

- Choose **Collection** for the action.

Open the Service editor for the second file system adapter, and choose **FS_Test** for the configuration.

- Choose **Extraction** for the action.

Important!

Some fields can only be configured in the Graphical Process Modeler such as concatenate and sort by filename or modification date.

(Continued on next page)

Exercise 7–1 File System Adapter

(Continued)

Create a Business Process (Continued)

5. Click **File > Save As**, and save the business process as **TestFSAdapter**. If you are asked to validate on save, then validate the business process.
6. Copy file1.txt from Desktop/Data files/Class Labfiles to **/home/student/fscoll** command line.

Note: You can use Filezilla to transfer the file. There is a preconfigured connection for root and student using the arrow next to the QuickConnect button. Choose the root user connection.

(Continued on next page)

Instructor Note: Help the students to upload the file from desktop to the command line using FileZilla. You can use the following login details to connect to the command line: Host: 192.168.40.100, Username: root, Password: password. and Port: 22

Exercise 7–1 File System Adapter

(Continued)

Check in and Test the Business Process

You can follow the steps that are listed to check in and test the business process.

1. Open the Admin console and choose **Business Process > Manager**.
2. Click **Go!** next to Create Process Definition.
3. Enter **TestFSAdapter** for the Process Name.
4. Select the **Check in Business Process created by the graphical modeling tool** radio button.
5. Click **Next**.
6. Click **Browse** and go to your saved business process. Select it and click **Open**.
7. Use **File system collect and extract** for a description and click **Next**.
8. Use the default settings for the remaining parameters. Make sure that the Business Process is enabled when you click **Finish**.
9. Search for your new business process, **TestFSAdapter** from the Business Process Manager screen.
10. Click **Execution Manager** when you get the results screen.
11. Choose **execute** in the execution manager screen and click **Go!**.

Note: No need to give a file when clicking Go! because the FSA will collect a file.

► Execute Business Process

Name: TestFSAdapter Instance ID: 7546 User: admin

Completed
Status: Success

Step	Service	Status	Advanced Status	Started	Ended	Status Report	Document	Instance Data
2	FS_Test	Success	None	04/13/2016 5:16:04 PM EDT	04/13/2016 5:16:04 PM EDT	i info	i info	i info
1	FS_Test	Success	1 file(s) collected.	04/13/2016 5:16:04 PM EDT	04/13/2016 5:16:04 PM EDT	i info	i info	i info
0	INITIATING_CONTEXT from user 'admin'	Success	None	04/13/2016 5:16:04 PM EDT	04/13/2016 5:16:04 PM EDT	None	None	None

* Inline Invocation

Bootstrapping

Introduction

A business processes can be scheduled to run at a fixed date and time. However, the trigger for a business process may be the arrival of data from outside Sterling B2B Integrator, such as a JMS queue. Bootstrapping is defined as the act of initiating a business process. If the Sterling B2B Integrator Scheduler bootstraps a business process, no outside activity is needed, although the business process may, and probably will, make use of external systems in the course of its operation. Some adapters can be configured to bootstrap a business process when they receive data.

Here are some additional facts about bootstrapping:

1. There can be a many-to-many relationship between bootstrapping adapters and business processes. That is, an adapter which bootstraps can initiate more than one business process.
2. Many adapters can be configured to start the same business process.
3. The business process that is bootstrapped can be selected dynamically. That is, the bootstrapping process can examine data, or use other criteria, to determine which business process to bootstrap.
4. The bootstrapping adapter can be at the beginning of a business process or can occur in the middle of a business process.
5. When creating a service to bootstrap, a system generated business process is built that uses the service by way of the Scheduler. The business process is normally named `Schedule_<service_configuration_name>`.

(Continued on next page)

Bootstrapping

(Continued)

Introduction (Continued)

Example:

Consider that a purchase order is being processed.

1. Someone needs to approve the purchase order.
 2. The order is sent to the approver and the business process waits for the order to return.
 3. The approver receives the order, approves, or disapproves it, and sends it back to Sterling B2B Integrator.
 4. When it receives the order, Sterling B2B Integrator is designed to be returned to the business process at the point where it was sent out so that it may continue on to the next step in the business process.
-

Exercise 7–2 Modify the File System Adapter

Instructions

You are required to complete the following activity to modify the File System Adapter to schedule a bootstrapping collection:

1. Click **Deployment > Services > Configuration** on the Admin console.
2. Search for the File System Adapter you configured earlier in this lesson. (In this example, search for **FS_Test**)
3. Click **edit** to make changes to the adapter configuration.
4. Type an appropriate description on the Name screen. Do not create or select a group. Click **Next**.
5. On the Collection screen, specify the appropriate settings for collecting data.
 - a. The Collection folder: parameter is mandatory. It is the name of the directory from which you want to collect files. Verify /home/student/fscoll
 - b. Leave the filename filter blank.
 - c. Collect files from subfolders within and including the collection folder: No
 - d. Use the absolute file path name for the document name: No
 - e. Start a business process once files are collected: **Yes**
 - f. Click Next.
6. Select **BasicInventoryProcess** from the drop-down menu on the Business Process screen, which you created in the previous exercise and click **Next**.

(Continued on next page)

Exercise 7–2 Modify the File System Adapter (Continued)

Instructions (Continued)

7. On the Schedule Type screen, choose **Run based on timer** option.
8. On the Schedule Settings screen, choose to run every 0 hours, 1 minute and click **Next**.
9. Accept the default settings at both Schedule Exclusions and Date Exclusions screens.
10. On the Extraction screen, verify **/home/student/fsext** in the Extraction folder: box and select Assign a specific name from the File naming convention radio buttons.
11. Click **Next**. The Extraction: User defined screen opens.
12. Type **%^Myoutput.txt** and click **Next**.
13. Review your selections on the Confirm screen
14. Click **Finish**.
15. Copy either Inventoryof xml file from datafiles to /home/student/fscoll
16. Allow one or two minutes for processing: From the **Admin Console**, select **Business Processes > Monitor > Current Processes**.
17. Monitor for a **BasicInventoryProcess** execution.
18. Click the Process ID number for the process. Verify that step 0 is your FSA.

Business Process Detail

Name: [BasicInventoryProcess](#) Instance ID: 8236 Status: Success State: Completed User: admin
 Deadline: None Contract ID: None

Action:

Steps 1-5 of 5

Step	Service	Status	Advanced Status	Started	Ended	Status Report	Document	Instance Data
0	FileSystemAdapter-FS_Test	Success	None	04/14/2016 11:28:24 AM	04/14/2016 11:28:24 AM	None	info	info
1	XML Encoder Service	Success	None	04/14/2016 11:28:24 AM	04/14/2016 11:28:24 AM	None	info	info
2	Assign Service	Success	None	04/14/2016 11:28:24 AM	04/14/2016 11:28:24 AM	None	info	info
3	Decision Engine Service	Success	1	04/14/2016 11:28:24 AM	04/14/2016 11:28:24 AM	None	info	info
4	Assign Service	Success	None	04/14/2016 11:28:24 AM	04/14/2016 11:28:24 AM	None	info	info

Exercise 7–3 Edit the File System Adapter

- Scenario** You need to edit the File System Adapter so it doesn't bootstrap and stop the scheduled execution. If a FSA is set to bootstrap it will not error on "No files to Collect".
1. Go to **Deployment > Schedules** in the UI.
 2. Accept the default of "**All**" in the last pane that is titled, "List Schedules", and click **GO!**
 3. Disable the Schedule by unchecking "**Enabled**" next to **FS_Test** in the list.
 4. Click **Deployment > Services > Configuration** on Admin console.
 5. Search for the **File System Adapter** you configured earlier in this lesson. (In this example, search for **FS_Test**)
 6. Click **edit** to make changes to the adapter configuration.
 7. Type an appropriate description on the Name screen. Do not create or select a group. Click **Next**.
 8. Specify the appropriate settings for collecting data on the Collection screen:
 - a. The Collection folder: parameter is mandatory. It is the name of the directory from which you want to collect files. Verify **home/student/fscoll**.
 - b. Leave the filename filter blank.
 - c. Collect files from subfolders within and including the collection folder: **No**
 - d. Use the absolute file path name for the document name: **No**
 - e. Start a business process once files are collected: **No**
 - f. Click **Next**.
 9. Review your selections on the Confirm screen and click **Save** and **Finish**.
-

OnFault Processing

Introduction

The onFault element is used to handle errors. You can include onFault elements in any complex activity for which it may be necessary to recover from faults so that the process can continue. The onFault element contains a fault handling activity. In a business process, a fault can occur in a complex activity that is not handled by an onFault element at that level of the process. When the fault code does not match the onFault element at a level, Sterling B2B Integrator begins to look for the correct onFault element. When no code attribute exists and no match to a defined attribute exists for an onFault element at a level, Sterling B2B Integrator moves to a general onFault (no code attribute) element. If it doesn't find a fault code, Sterling B2B Integrator halts the process.

OnFaults are commonly tied to sequences. This could be a sequence that covers the entire business process flow or a smaller sequence such as a choice or repeating loop. Each onFault can also be setup to trigger on different error conditions. This could be specific error or setup to be generic and trigger on any error.

For each onFault condition you build a subroutine to run if an error condition is met. It is common to build some type of alert bp like email people or a file drop off bp and then invoke those bp's through the subroutine. This way you can reuse common bp's instead of building the steps into every onFault.

SMTP Send Adapter

Overview

The SMTP Send Adapter allows documents to be sent to any valid email address using an accessible (SMTP-aware) mail server. It does this by enabling Sterling B2B Integrator to mail (send) documents using SMTP, within Sterling B2B Integrator, to the designated mail server. The mail server system examines the document and does further processing, including the actual sending of the email.

This adapter can be used when messages and documents need to be sent during the execution of a business process.

How the SMTP Send Adapter Works

The SMTP Send Adapter can mail (send) documents to the mail server of trading partners. The document is then mailed to the recipient designated in the To field of the configuration. This adapter picks up the document to be sent from the primary document in the business process.

Exercise 7–4 OnFault Processing Using SMTP Send Adapter

Scenario The SMTP Send Adapter is designed to send a file as an email attachment. If an email notification of No Files to Collect is required, we must capture the Previous Not Success Advanced Status, which is issued from the SMTP adapter. The DOMToDoc XPath expression retrieves this notification from process data and writes into the PrimaryDocument to give the required input file to the SMTP Send Adapter.

Note: A valid mail host name and email id is provided to create business process as required in this exercise.

Setting the OnFault Service

In order to use your updated configuration of the File System Adapter, inside the Graphical Process Modeler, click **View > Refresh Services** to get an updated configuration for the FSA instance in the business process.

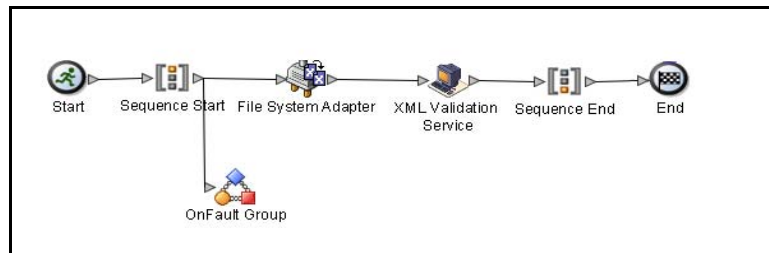
1. Open the **Graphical Process Modeler**.
2. Create a business process that is comprised of the following:
 - (1) Start
 - (1) End
 - (1) Sequence Start
 - (1) Sequence End
 - (1) File System Adapter
 - (1) OnFault Group
 - (1) XML Validation Service

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Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Setting the OnFault Service (Continued)

3. Arrange and connect the services as shown in the following figure.



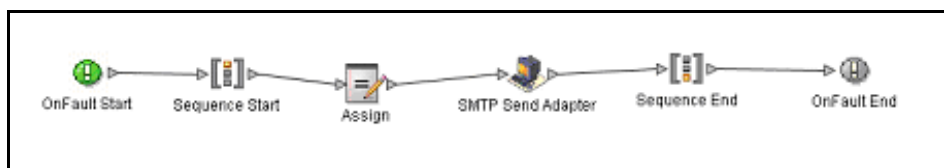
4. Click the **File System adapter** and configure as follows:
- Select **FS_Test** from the Configuration drop-down menu
 - Set the Action parm to **Collection**.
5. Click the **XML Validation Service** and select **XML Validator_WellFormed** from the Configuration drop-down menu.
6. Click the **OnFault Group** activity.
- Type **No files to collect** under the Error Value parm (must match).
 - Rename Default to **First Fault**.
 - Select **add** to create a second fault. Under the name type **Second Fault**.
 - Leave blank under error value (this will be an error catch-all.)
7. Locate and click the **First Fault** in the Navigation pane.
- Between the OnFault Start and OnFault End, add
 - (1) Assign
 - (1) SMTP Send Adapter
 - (1) Sequence Start/End

(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Setting the OnFault Service (Continued)

8. Arrange and connect the services as shown in the following figure.



- Click the **Assign** under the value for append type false.
- Under the value for from type:

`DOMToDoc (Prev_NotSuccess_Adv_Status, 'PrimaryDocument')`

- Under value for to type '.' (just a period no quotes).

Important!

The SMTP Send Adapter is designed to send a file. Since there is no input file to send we must capture the Previous Not Success Advanced Status of No files to Collect from process data and write into PrimaryDocument to give the SMTP Send Adapter the required input file. DOMToDoc is an engine call that moves data from ProcessData (DOM) to Primary Doc (Doc). The period (.) means current Primary Doc. So steps b and c say copy everything in Process Data under tag Prev_NotSuccess_Adv_Status and copy to Primary Doc.

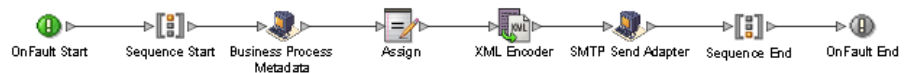
(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Setting the OnFault Service (Continued)

- d. Click the SMTP Send Adapter, select the SMTP Send Adapter from the Configuration drop-down menu.
 - e. Type the valid host name **192.168.40.1** under xport-smtp-mailhost.
 - f. Type your email address **student01@ibm.com** for xport-smtp-mailto.
9. Locate and click the **Second Fault** in the Navigation pane.
- a. Between the OnFault Start and OnFault End, add;
 - Sequence start/end (1)
 - Business Process Metadata service
 - Assign
 - XML Encoder service
 - SMTP Send Adapter

Link together as shown in the following figure.



(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Setting the OnFault Service (Continued)

- b. Click Business Process Metadata service and choose **BPMetaDataInfoService** as the configuration and accept the default values for all other parameters.
- c. Set the properties as follows on the Assign:

Name	Value
To	Server
From	substring-before(//WFC_ID/text(),':')

- d. Choose **XMLEncoder** on the XML Encoder as the configuration and set the properties as follows:

Parameter	Value
Mode	Create Document using xPath
OutputToProcessData	Yes
root_element	ErrMsg
xPath	//ProcessData/* Note: Do not select the xpath check box. This field is looking for an xpath statement.

(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Setting the OnFault Service (Continued)

- e. Choose **SMTP Send Adapter** as the configuration on the SMTP Send adapter, and set the properties as follows:

Parameter	Value
Xport_SMTP_mailhost	192.168.40.1
Xport_SMTP_mailport	25
Xport_SMTP_mailto	student01@ibm.com
Xport_SMTP_mailsubject	concat('Process #',//WORKFLOW_ID/text(),' on server ',//Server/text(),' has encountered an error') Note: Type this all on one line and click “use xPath”.

- f. Save and validate this business process name the process **onFault_test**.

Important!	The XPath statement that you entered for the subject line concatenates literal values that are placed in single quotes, and variable values from ProcessData. WORKFLOW_ID is an element that is created in ProcessData by the BP Metadata service, and contains the process instance id of your execution. The tag Server was created by an XPath statement in the Assign. That statement searched a tag in ProcessData named WFC_ID that was created by the Error service. The substring-before statement located a colon in the value and extracted everything before that byte. Since the data held before the colon is the name of the server Sterling B2B Integrator is running on, the value extracted will be the server name.
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(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Check in and Test the Business Process

You can check in and test the business process in the following manner.

1. Open the Admin console and choose **Business Process > Manager**.
2. Click **Go!** next to Create Process Definition.
3. Enter **OnfaultSMTP_test** for the Process Name.
4. Make sure that the radio button titled **Check in Business Process created by the graphical modeling tool** is selected.
5. Click **Next**.
6. Click **Browse** and go to your saved business process. Select it and click **Open**.
7. Type **Checking onfault using SMTP** for the description and click **Next**.
8. Use the default settings for the remaining parameters. Make sure that the Business Process is enabled and click **Finish**.

(Continued on next page)

Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Testing the Business Process

You are required to run the business process three different times and follow the steps that are listed to test the business process.

Receive Email from First Fault

- Execute the business process without an input file.
- You will receive an email containing message with no files to collect.

Receive Email from second Fault

- Copy file1.txt to /home/student/fscoll and execute the business process.
- You will receive an email containing the Process Data.

Receive No Email

- Copy inventoryOf3.xml to /home/student/fscoll and execute the business process.
- The business process will complete without any error and you will not receive any email.

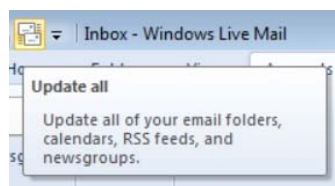
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Exercise 7–4 OnFault Processing Using SMTP Send Adapter (Continued)

Viewing Emails in Windows Live Mail

You can follow the steps to view the emails in the Outlook Express.

1. Open Windows Live Mail by going to **Start > All Programs > Windows Live Mail**.
2. Enter the username as **student1@ibm.com**.
3. Enter the password as **password1**.
4. Click **Update all** in the upper left corner to check for the emails.



Sample Email



Command Line 2 Adapter

Overview

The Command Line 2 adapter is a second-generation adapter that allows Sterling B2B Integrator to run a program from a command line in a business process. This includes executable programs, scripts, or operating system (OS) commands external to Sterling B2B Integrator.

The Command Line 2 adapter operates only in a remote implementation. This does not necessarily mean that it must run remotely. The Command Line 2 adapter runs in a separate JVM (Java Virtual Machine), which may be on the computer where Sterling B2B Integrator is installed or on a remote computer. The Command Line 2 adapter supports large files up to 12 GB and provides better memory allocation than the Command Line adapter. The Command Line 2 adapter will eventually replace the Command Line adapter.



Caution

Use the Command Line 2 adapter only for non-interactive programs that run from the command line. This adapter is not intended to work with browser-based or other Graphical UI-based programs. Such programs introduce an interactive element into a business process, which is typically setup to be automated.

Use the Command Line 2 adapter to invoke a program that:

- Encrypts and decrypts data that you want to send or receive securely over the Internet
- Manipulates data, such as change every occurrence of one letter to another
- Pages someone
- Initiates a business process
- Initiates a remote system

These are just a few examples out of many possible uses.

(Continued on next page)

Instructor Note: Inform students to change the path name in the scripts according to their folder structure using Vi Editor or opening in notepad. Guide the students to modify the script.

If the script is not running you can try to convert the file for Unix in Gedit as CR/LF my have been inserted by Windows. Make sure that the students have replaced the right path in the scripts.

Command Line 2 Adapter

(Continued)

How the Command Line 2 Adapter Works

The following steps summarize how the Command Line 2 adapter is typically used in a business process:

1. The adapter writes the content of the current primary document to a file in the working directory that is specified as the value of the working directory parameter. The name of this file is specified by the value of the inputFile parameter.
2. Sterling B2B Integrator runs an executable program that picks up the file and sends it to the legacy system.
3. The legacy system returns a file, which now includes the customer billing information, and the adapter retrieves it. The file that is returned is specified by the value of the outputName parameter.
4. The adapter reads the file contents into the primary document.
5. Sterling B2B Integrator performs the next operation in the business process.

Implementing the Command Line 2 Adapter

You can implement a Command Line 2 adapter to do the following:

- Execute commands using the command line from within a business process.
- Invoke the Command Line 2 adapter on a schedule and then start a new business process using the output from the adapter.

Important!	This can be used if you want to schedule a command-line program that accesses a legacy database on a regular schedule and then uses the output in a business process.
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Command Line 2 Adapter

(Continued)

Before You Begin

Before you begin to implement the Command Line 2 adapter, complete the following tasks:

1. If running Sterling B2B Integrator version 5.2.4.1 or later you have to enable the CLA2 adapter on the local machine. Prior versions CLA2 starts as part of startup script. (Remote installs are unchanged)
 1. Edit *Sandbox.cfg* in the properties directory by adding the line:
LAUNCH_CLA2_SERVER=true
 2. Shutdown B2Bi and run setupfiles.sh
2. Create and test the command-line program or command to make sure that it works.
3. Determine the working directory where you will be processing your commands.

Process Overview

To implement the Command Line 2 adapter, complete the following tasks:

1. Create a Command Line 2 adapter configuration.
2. Configure the Command Line 2 adapter.
3. Create and enable a business process that includes the Command Line 2 adapter.

Important!	If you are configuring a Command Line 2 adapter to start a business process, create the business process before configuring the adapter.
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4. Test the business process and the adapter.
 5. Run the business process.
-

Exercise 7–5 Command Line 2 Adapter

Instructions

For ease of understanding, the instructions are chunked into three parts that are as listed:

- Create the Service Configuration
- Create a Business Process
- Check in and Test the Business Process

Create the Service Configuration

You can create the service configuration in the following manner.

1. Open **copy.sh** (found in Datafiles) and replace the copy command with the following command:

```
cp /home/student/fscoll/Trigger.XML /home/student/fsext
```

Note: Verify whether the path in the command is according to your folder path.

2. Upload the copy.sh script to **/home/student** and **Trigger.xml** file to **/home/student/fscoll**.
3. Log in to admin console.
4. Click **Deployment > Services > Configuration** on Admin console.
5. Click **Go!** in **New Service**.
6. Type in **Command Line Adapter 2** and click **Next**.
7. Enter the following required information, and click Next.
 - a. Enter **CLA2_Test** for Name.
 - b. Enter **Command Line 2 Test** for Description.
 - c. Click **None** for Select a group.

(Continued on next page)

Instructor Note: There are two scripts; copy.sh, and list.sh. The steps for configuring command line 2 adapter uses copy.sh. If you have enough time, you can ask the students to try configuring the list.sh script too. You can also ask the students to create a script to rename the existing file and configure it on command-line adapter.

Exercise 7–5 Command Line 2 Adapter (Continued)

Create the Service Configuration (Continued)

8. Enter the following required information about the Services Configuration Properties screen, and click **Next**.
 - a. Enter **localhost** for Remote Name.
 - b. **9052** for Remote Port.
 - c. Enter **/home/student/copy.sh** for Command Line.
 - d. Enter **/home/student/** for Working Directory.
 - e. Leave Authentication active along with default certificate.
 - f. Choose **No** for Turn on debugging messages.
 - g. Click **Yes** for Wait on process to continue?
 - h. This service does NOT invoke a business process. Click **Next**.
9. The command-line process DOES NOT require an input file. Click **Next**.
10. DO NOT use the output generated by the command-line process.
11. Choose **Do not use schedule** and click **Next**.
12. Ensure that the service is enabled for business processes and click **Finish**.

Note: Similarly, you can configure Command Line 2 adapter for list.sh or script that you want to execute through business process.

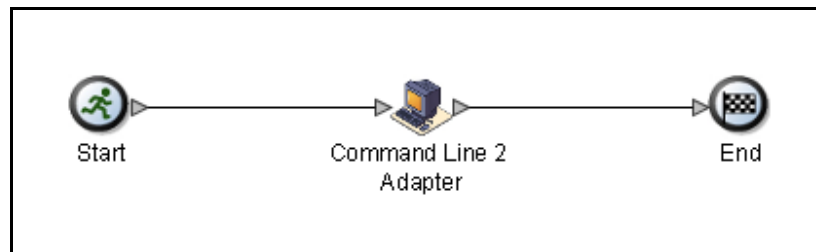
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Exercise 7–5 Command Line 2 Adapter (Continued)

Create a Business Process

You can create a business process in the following manner.

1. Open the Graphical Process Modeler. If it is already open, click **View > Refresh Services**.
2. Click **File > New**.
3. Drag a **Start**, **End**, and **Command Line 2 Adapter** into the workspace and connect them as shown.



4. Double-click the **Command Line 2 Adapter** to open its properties editor.
5. Choose **CLA2_Test** from the **config** drop-down box.
6. Save and validate your business process. Name it **CLA2_Test**

(Continued on next page)

Exercise 7–5 Command Line 2 Adapter (Continued)

Check in and Test the Business Process

You can follow the steps that are listed to check in and test the business process.

1. Open the admin console and choose **Business Process > Manager**.
 2. Click **Go!** next to Create Process Definition.
 3. Name the process **CLA2_Test**.
 - a. Make sure that the **Check-in Business Process created by the graphical modeling tool** radio button is selected.
 - b. Click **Next**.
 4. Click **Browse** and go to your saved business process. Select it and click **Open**.
 5. Give it a suitable description and click **Next**.
 6. Use the default settings for the remaining parameters. Make sure that the business process is enabled when you click **Finish**.
 7. Search for your new business process from the Business Process Manager screen.
 8. Click **Execution Manager** when you get the results screen.
 9. Choose **execute** in the execution manager screen. There is no input file.
 10. Click **Go!**
 11. Monitor the progress of the business process and after successful completion of business process the Trigger.xml file is copied to the fsxt folder.
-

The Invoke Business Process Service

Overview	The Invoke Business Process Service is used to start an instance of a subprocess (child) from within the current instance of the business process (parent). Note: The Invoke Business Process Service is also referred to as the Invoke Sub-process Service.
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How the Invoke Business Process Service Works	The Invoke Business Process Service, when used, instantiates an instance of a subprocess, passing the current workflow context and the primary document to the new instance. ■ When the Invoke Business Process Service is set to synchronous mode, the parent suspends processing until it receives data from the child. In synchronous mode, the parent is notified when the child encounters errors. ■ When the Invoke Business Process Service is set to asynchronous mode, the parent and child process data simultaneously and independently of each other. Therefore, the parent does not receive notification when the child encounters errors. ■ When the Invoke Business Process Service is set to run a subprocess inline, the subprocess runs as part of the parent process, sharing the same process data. ■ When the Invoke Business Process Service is set to run in embedded mode, the subprocess runs with no persistence, meaning that no record of the process is recorded in Sterling B2B Integrator and no tracking is done.
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The Invoke Business Process Service (Continued)

Implementing the Invoke Business Process Service

To implement the Invoke Business Process Service, complete the following tasks:

1. Create and enable a business process that includes the Invoke Business Process Service.
2. Test the business process and the service.
3. Run the business process.

Important!	The Invoke Business Process service does not require you to build a configuration as all of the configuring is done in the GPM/BPML. If you do build a configuration all that is entered is a name and description. But it can then be used as any other configuration. Commonly developers just use the pre-built InvokeBusinessProcessService configuration and configure in the GPM/BPML.
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The Invoke Business Process Service (Continued)

**Graphical Process
Modeler
Configuration**

The following table describes the fields that are used to configure the Invoke Business Process Service in the Graphical Process Modeler:

Important!	The names in parentheses represent the corresponding field names in the Graphical Process Modeler. This information is provided for your reference.
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Field	Description
Config	Name of the service configuration.
COPY_SERVICE_PARAMS	Invoke Business Process Service parameters passed to the subprocess. Valid values are True (default) and False.
Business Process Definition Name (WFD_NAME)	Business process that is used in the service configuration. Valid values include all the business processes installed.
PASS_STATUS_REPORT	Pass status report to parent. Valid values are: ■ ON_SUCCESS ■ ON_ERROR ■ ALWAYS ■ NEVER Default is ON_ERROR. Required.
Continued on next page	

The Invoke Business Process Service (Continued)

Graphical Process Modeler Configuration (Continued)

Field	Description
Invoke Mode (1=Async, 2=Sync) (INVOKE_MODE)	Mode in which to run the subprocess. Optional. Valid values are: ■ async = asynchronous (default) ■ sync = synchronous ■ inline = In-line ■ embedded = starts business process in SYNC mode with Enable Transaction on, and runs the subprocess in the same transaction as the Invoke Sub-Process service. The subprocess is run with no persistence. Note: When persistence level is error only, sync invoke mode is not supported.
List of ParmS to override (PARM_LIST)	Parameters to override
Type of Error On Which to Notify Parent Process (NOTIFY_PARENT_ON_ERROR)	Errors the subprocess reports to the parent business process (synchronous mode only). Valid values include: ■ Continue parent only if a service-generated error occurs in the subprocess. ■ Continue parent only if a system-generated error occurs in the subprocess. ■ Continue parent if any type of error occurs in the subprocess. ■ Halt subprocess if any error occurs (for potential correction and resume). This option does not notify the parent of the error from the subprocess, and the subprocess is halted.

Exercise 7–6 Invoke Business Process Service

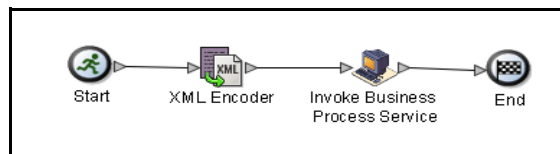
Instructions For ease of understanding, this procedure is chunked into two parts that are as listed:.

- Create a Business Process
- Check in and test the business process

Create a Business Process

You can create a business process in the following manner.

1. Open the Graphical Process Modeler. If it is already Open, click **View > Refresh Services**.
2. Click **File > New**.
3. Drag a **Start**, **End**, an **XML Encoder Service** and an **Invoke Business Process Service** into the workspace and connect them as shown.



4. Double-click on the **XML Encoder** to open its properties editor.
5. Choose **XMLEncoder** from the config drop-down box.
 - Mode: Use existing XML document
 - Output_to_process_data: Yes

(Continued on next page)

Exercise 7–6 Invoke Business Process Service (Continued)

Create a Business Process (Continued)

6. Double-click the **Invoke Business Process Service** to open its properties editor.
7. Choose Invoke Business Process Service from the config drop-down box.
8. Specify the following values in the Properties Editor:
 - Invoke_mode: Async
 - Wfd_name: Select your Basic Inventory process.
9. Save the file as **INVOKE_Test**.

(Continued on next page)

Exercise 7–6 Invoke Business Process Service (Continued)

Check in and Test the Business Process

You can check in and test the business process in the following manner.

1. Choose **Business Process > Manager**, and open the admin console.
2. Click **Go!** next to Create Process Definition.
3. Name the process **Async**.
 - a. Make sure that the **Check-in Business Process created by the graphical modeling tool** radio button is selected.
 - b. Click **Next**.
4. Click **Browse** and navigate to your saved business process. Select it and click **Open**.
5. Give an appropriate description. Click **Next**.
6. Use the default settings for the remaining parameters. Make sure that the Business Process is enabled when you click **Finish**.
7. Search for your new business process from the Business Process Manager screen.
8. Click **Execution Manager** when you get the results screen.
9. Choose **execute** in the execution manager screen.
10. Click **Browse** and choose the **InventoryOf11.xml**.
11. Click **Go!**
12. Monitor the progress of the business process.

(Continued on next page)

Lightweight JDBC Adapter

Overview

The Lightweight Java Database Connectivity (JDBC) adapter allows you to retrieve data from a JDBC-compliant database or update a JDBC-compliant database as part of a business process within Sterling B2B Integrator, using one of the following methods:

- XPath queries for dynamic SQL queries (where the definition of the SQL statement is not known when the business process is defined)
- Constants, when lookups are not necessary

How is the Lightweight JDBC Adapter Different?

Unlike the JDBC adapter, the Lightweight JDBC adapter does not use a map, where the SQL statements are fixed. The Lightweight JDBC adapter allows you to specify parameters in the SQL statement that are supplied at run time.

The Lightweight JDBC adapter executes a query, and you receive an XML document that contains the results. Then you can either:

- Load the XML document into process data.
- Carry the XML document forward into another service in the business process.

Use the Lightweight JDBC adapter if you simply want to get data, and need the flexibility of specifying various SQL queries. Use the JDBC adapter if you plan to always execute the same SQL statement, or if you have complex output and you want to manipulate the data from a database and control the structure of the output.

Important!	The Lightweight JDBC adapter processes only one result set if you script multiple queries to run in stored procedures.
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Lightweight JDBC Adapter

(Continued)

How the Lightweight JDBC Adapter Works

The Lightweight JDBC adapter can start a business process, or it can be used in the middle or at the end of a business process. This service can be scheduled to run at weekly or timed intervals.

The following steps summarize how the Lightweight JDBC adapter might be used in a business process:

1. Using the parameters that are specified in the Graphical Process Modeler for the Lightweight JDBC adapter configuration, the adapter executes a query.
2. The adapter receives the query results from an SQL database query.
3. The adapter generates a primary document that contains the query results in XML format.
4. The adapter sends the query results to the next step in the business process.

Setting Up the Lightweight JDBC Adapter

To use the Lightweight JDBC adapter, you must complete this process:

1. Set up a connection to an external database.
2. Create and enable a Lightweight JDBC adapter configuration.
3. Create a business process using the Lightweight JDBC adapter, and enable it.

Important!	If you want to use XPath queries, you will first need to load the data these queries will reference into process data. For example using the XML Encoder service or DocToDOM in your business process.
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Lightweight JDBC Adapter

(Continued)

Setting Up the Lightweight JDBC Adapter (Continued)

4. Double-click the **Lightweight JDBC Adapter** icon in the Graphical Process Modeler, and in the Service Editor, specify the parameters or queries the adapter configuration will use.
5. Test the business process and the adapter.

Creating a Lightweight JDBC Adapter Configuration

To create a Lightweight JDBC adapter configuration:

1. Select **Services > Configuration** from the Sterling B2B Integrator Deployment menu.
2. Click **Go!** under Create, next to New Service.
3. Type **Lightweight JDBC Adapter** in the Service Type dialog box and click **Next**.
4. Complete the fields specific to the adapter and click **Finish**.
5. Specify additional parameters using the Service Editor in the Graphical Process Modeler.

Code Lists

JDBC connections between Sterling B2B Integrator and outside databases are very powerful, useful tools. However, creating that connection can be complicated. For some tasks, you may not need such advanced functionality. In these cases, you can make use of the Sterling B2B Integrator function; the code list.

Code lists are created as tables through the Sterling B2B Integrator UI, but are held as records within the Sterling B2B Integrator database, in a table named `codelist_xref_item`. These records can be accessed through JDBC adapters in your business process, and also through maps you create. Like any other table, you can select, update, and insert records, and call stored procedures or functions when working with code lists.

Lightweight JDBC Configuration

Introduction

The following table describes the fields that are used to configure the Lightweight JDBC adapter in Sterling B2B Integrator:

Important!	The field names in parentheses represent the corresponding field names in the Graphical Process Modeler. This information is provided for your reference.
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Field	Description
Name	Unique and meaningful name for adapter configuration. Required.
Description	Meaningful description for the adapter configuration, for reference purposes. Required.
Start a new business process (StartNewWorkflow)	Whether to start a new business process. Valid values are: ■ This Lightweight JDBC adapter configuration will start a new business process. ■ This Lightweight JDBC adapter configuration will not start a new business process.
Run As User	Specify the user that this schedule runs under. Click the icon next to the field for a list of users from which to select.
Do not use schedule	If the field is selected, this service does not start a business process and does not run on a schedule.
Run service based on timer every	Valid values are the hour and minutes at which to run the service. Indicate whether you want the service to run at startup.
Continued on next page	

Lightweight JDBC Configuration

(Continued)

**Introduction
(Continued)**

Field	Description
Run service daily at	Valid values are the hour and minutes at which to run the service, daily. Indicate whether you want the service to run at startup.
Run service weekly on	Valid values are the day of the week, the hour, and the minutes at which to run the service. Indicate whether you want the service to run at startup.
Business Process	Select the business process that will be started by this service.
Pool Name	Specify the database pool name. Optional.
XML Result Root Tag (result_name)	Root tag element. The name of the tags in the XML document for query results that will be attached to the process data by the adapter. Required.
XML Result Row Tag (row_name)	Row tag element containing all the column tags. The column tag names are not configurable. The tag names are generated by the column name returned in the result set. Required.
Continued on next page	

Lightweight JDBC Configuration

(Continued)**Introduction
(Continued)**

Field	Description
Query Type (query_type)	Result set or the number of rows affected by an action query returned by the SQL statement. Valid values are SELECT and ACTION. Required. ■ SELECT – Returns results. The rows affected by the query, if available, are returned into the row tag itself, for example, <root><row>5</row></root>. ■ ACTION – Returns the number of rows affected. For example, if you are updating records in a database, the number of rows updated is returned. ■ Stored procedure/function – Required if value in paramtype1 field is Cursor.
SQL Statement (sql)	Hard-coded SQL query that will query a database. It must be in valid SQL syntax. Optional.
Business process (InitialWorkflowId)	Business process you want the Lightweight JDBC adapter to start, if any. Required if you selected This Lightweight JDBC adapter configuration will start a new business process. Otherwise, optional.

(Continued on next page)

Lightweight JDBC Configuration

(Continued)

Graphical Process Modeler Configuration

The following table describes the fields that are used to configure the Lightweight JDBC adapter in the Graphical Process Modeler:

Field	Description
Config	Name of the service configuration.
param1 - param20	XPath query that searches the process data tree for a match. If defined as XPath, you must also select the XPath check box in the Service Editor of the Graphical Process Modeler for the parameter. Optional.
paramtype1 - paramtype20	Parameter type that corresponds to the parameter's number, for example, param1 and paramtype1. Every parameter specified must have the corresponding parameter type. Valid values are: ■ Character Stream ■ Cursor ■ Date ■ Double ■ Float ■ Integer ■ Long ■ String Cursor runs stored database functions. If specified, this must be paramtype1. If you want to pass any other parameters to the database stored function, those must start in the paramtype2 field.

Restrictions on Submitting Stored Functions

Overview

The following restrictions and guidelines apply when submitting stored functions to a database using the JDBC adapter:

- If the stored procedure is a procedure and not a function, the first argument to the procedure must be an output parameter. Even if the procedure has no data to return, it must declare the first argument as an output Cursor.
- The output argument must be in the first position.
- The first parameter type must be Cursor type.
- You must provide a value for the first parameter, even though it is not used.
- The Query type must be declared as Stored Procedure/Function type.
- The SQL query must be defined as an anonymous block with a begin and end statement.
- When defining the Integer data type using the Graphical Process Modeler (GPM), selecting Integer will produce Double in the BPML. If you need an Integer type, edit the BPML through the Sterling B2B Integrator interface and replace Double with Integer.

Important!	If you edit the file at a later time using the Graphical Process Modeler, this edit overwrites your earlier changes and sets the value back to Double.
-------------------	--

Exercise 7–7 Creating a Code List

Scenario The ERP system generates XML files that must be routed to different business processes within Sterling B2B Integrator. The files are all placed into a single directory on disk. The required processes are in place, but must create a method for dynamically selecting those processes, which are based on the file name. The process names are stored into a table (code list) and retrieve them during processing, invoking them as appropriate.

Instructions For ease of understanding, this procedure is chunked into two parts:

- Create the code list
- View the code list

Create the Code list You can create a codelist in the following manner.

1. Go to **Trading Partner > Code Lists** from the Sterling B2B Integrator UI.
2. Click **GO!** to create a new codelist.
3. Name your new codelist **SelectProcess**. Leave Sender and Receiver Identities blank.
4. Click **Next**.

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Exercise 7–7 Creating a Code List

(Continued)

Create the Code list (Continued)

- Click **Add New Code**.
- Fill the first set of parameters given in the following table and click **Save**. Repeat the step 5 and fill other two set of parameters in the table.

Sender Code	Receiver Code	Description
InventoryOf11.xml	BasicInventoryProcess	Qty check for 15 process
InventoryOf3.xml	BasicInventoryProcess1	Qty check for 10 process
Trigger.xml	CLA2_Test	Command Line Adapter 2 process

Note: Ensure that the business processes entered in the Receiver Code are checked in and the exact name is typed as it is case-sensitive.

- When you have finished filling in the codes, click **Next**, and then **Finish**.

Important	Ensure that you delete all Non-default versions of the code list if you have to make edits. The LWJDBC can't read default flags like the GUI can and will give all versions as results.
------------------	---

Viewing the Codelist

You can view the codelist as it appears in the DB by following the steps that are listed.

- Go to **Operations > System > Support Tools > SQL Manager**.
 - Type **select * from codelist_xref_item where list_name = 'SelectProcess'** in the SQL Manager window.
 - Click **execute**.
-

Exercise 7–8 Lightweight JDBC Adapter with Codelist

Scenario	You are required to create a business process with Lightweight JDBC Adapter which verifies the input file process name with the receiver code entered in code list and proceeds further in invoking the business process. This functionality will provide security that the input file with different process name cannot be processed and executed.
-----------------	--

Instructions	For ease of understanding, this procedure is chunked into two parts that are as listed: ■ Configuring Lightweight JDBC Adapter ■ Creating a Business Process
---------------------	---

Configuring the Lightweight JDBC Adapter

You can configure the Lightweight JDBC adapter in the following manner.

1. Log in to admin console.
2. Click **Deployment > Services > Configuration** on admin console.
3. Click **Go!** under New Service.
4. Click **Lightweight JDBC Adapter**, and click **Next**.
5. Use **LWJDBC_Test** for a name, and **LWJDBC adapter** for a description, and click **Next**.
6. Accept the default settings and make sure **Will not start a new Business Process** is selected. Click **Next**.

(Continued on next page)

Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Configuring the Lightweight JDBC Adapter (Continued)

7. Specify the following values in the Parameters screen:

Property	Value
Pool Name	DB2pool
XML Result Root Tag	Document
XML Result Row Tag	Query
Query Type	(Leave Blank)
SQL Statement	(Leave Blank)

Note: If Oracle database is used in the Sterling B2B Integrator instance then you will get an option to choose oraclePool.

8. Click **Next**.
9. Ensure that the service is enabled for business processes and click **Finish**.

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Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process

You can create a business process by following the steps that are listed:

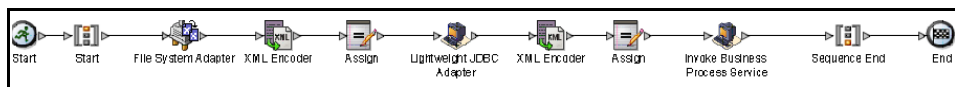
1. Open the Graphical Process Modeler. If it is already open, click **View > Refresh Services**.
2. Select **File > New**.
3. Drag the following activities onto your workspace:
 - a. Start (1)
 - b. End (1)
 - c. Sequence Start/Sequence End (1)
 - d. Assign (2)
 - e. File System Adapter (1)
 - f. XML Encoder (2)
 - g. Lightweight JDBC Adapter (1)
 - h. Invoke Business Process Service (1)

(Continued on next page)

Exercise 7-8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process (Continued)

4. Connect the stencils as shown:



5. Click the **Sequence Start** and rename it as **Start**.
 a. Click the **File System Adapter** and set the properties as shown:

Property	Value
Configuration	FS_Test
Action	Collection
Collect Multiple	No
Delete After Collect	Yes

(Continued on next page)

Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process (Continued)

6. Click the first **XML Encoder** and set the properties as shown in the following table:

Property	Value
Configuration	XML Encoder
mode	Use Existing XML Document
output_to_process_data	Yes
exhaust_input	Yes

7. Click the first **Assign** and set the properties as shown:

Property	Value
from	//PrimaryDocument/@SCIOBJECTID
to	DocHolder

(Continued on next page)

Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process (Continued)

8. Click the **Lightweight JDBC Adapter** and set the properties as shown:

Property	Value
Configuration	LWJDBC_Test
param1	//FileName/text() (Click “use xPath”)
param2	SelectProcess (Do NOT click “use xPath”, this parameter will be used as a literal value)
paramtype1	string
paramtype2	string
query_type	select
sql	select RECEIVER_ITEM from CODELIST_XREF_ITEM where SENDER_ITEM = ? and LIST_NAME = ? (place this statement all on one line)

(Continued on next page)

Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process (Continued)

9. Click the second **XML Encoder** and set the properties as shown:

Property	Value
Configuration	XML Encoder
mode	Use Existing XML Document
output_to_process_data	Yes
exhaust_input	Yes

10. Click the second **Assign** and set the properties as shown:

Property	Value
from	//DocHolder/@SCIObjctID
to	PrimaryDocument

11. Click the **Invoke Business Process Service** and set the properties as shown:

Property	Value
Configuration	InvokeBusinessProcessService
INVOKE_MODE	Inline
WFD_NAME	string(//Document/Query/RECEIVER_ITEM) (Click “use xPath”)

(Continued on next page)

Exercise 7–8 Lightweight JDBC Adapter with Codelist (Continued)

Creating a Business Process (Continued)

-
12. Save and validate the business process with the name **LWJDBC_Test**.
 13. Check in the process in dashboard with the name **LWJDBC_Test**.
 14. From the UNIX command-line, copy **InventoryOf3.xml**, **InventoryOf11.xml** and **Trigger.xml** into the **/home/fscoll** directory.
 15. Navigate to **/home/student**. Verify and delete if you still have the **Trigger.xml** in fsext folder that was copied from the Command Line 2 Adapter exercise.
 16. Run your process 3 times, without input file.
-

Result:

The three business processes; BasicInventoryProcess, BasicInventoryProcess1 and CLA2_Test are executed based on the input file.

Wait Service

Overview	The Wait service allows a business process to wait for a period of time before moving on to the next step in the process. The Wait service does not hold its active thread during this wait time, which can reduce resource consumption.
How the Wait Service Works	The Wait service performs a function similar to the Sleep service: it allows you to pause a business process for a set length of time. However, unlike the Sleep service, the Wait service does not retain an active processing thread, which frees the thread for other processing jobs. Use the Wait service when you need to put a business process in a Wait state for at least one minute. If you need to put a business process in a wait state for less than a minute, use the Sleep service.
Implementing the Wait Service	A configuration of the Wait service is installed with Sterling B2B Integrator.

Exercise 7–9 Wait Service

-
- | | |
|-----------------|---|
| Scenario | <p>This exercise is designed to demonstrate:</p> <ul style="list-style-type: none">■ The use of the All Start and how Sterling B2B Integrator handles business process contexts during processing.■ The use of the Repeat activity, and how to properly construct a loop with an index.■ The use of the Wait Service to pause a process during execution. |
|-----------------|---|

-
- | | |
|---------------------|--|
| Instructions | <p>You can create the business process in the following manner.</p> <ol style="list-style-type: none">1. Open the Graphical Process Modeler, if it is not already open.2. Click New to start a new business process. |
|---------------------|--|
-

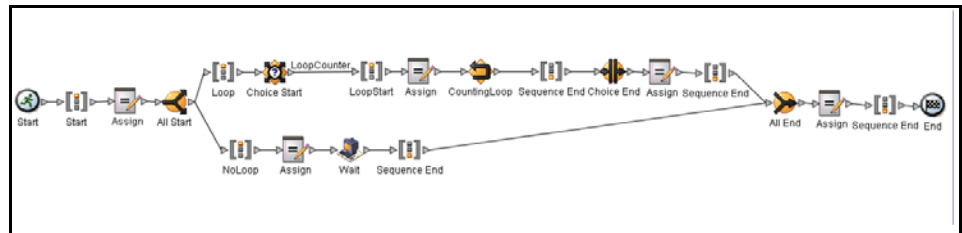
(Continued on next page)

Exercise 7–9 Wait Service

(Continued)

Instructions (Continued)

3. Move the following operations and services to your workspace by dragging the icons from the Stencil to the workspace:
 - (1) Start
 - (1) End
 - (4) Sequence Start
 - (4) Sequence End
 - (5) Assign
 - (1) All Start
 - (1) All End
 - (1) Choice Start
 - (1) Choice End
 - (1) Repeat Activity
 - (1) Wait
4. Connect the activities in the order as shown in the following figure.



(Continued on next page)

Exercise 7–9 Wait Service

(Continued)

Instructions (Continued)

5. Rename the Sequence Starts as follows:
 - a. First Sequence Start = **Start**
 - b. First Sequence Start in top branch of All Start = **Loop**
 - c. Second Sequence Start in top branch of All Start = **LoopStart**
 - d. Sequence Start in bottom branch of All Start = **NoLoop**
6. Click the first **Assign** in the process, and set the values as follows:
Constant = 0
To = LoopCount
7. Click the **Assign** in the bottom branch of the All Start, and set the values:
Constant = The NoLoop Branch has executed
To = NoLoop
8. Click the **Wait** in the process, and choose **WaitService** under **Config** drop-down list.
9. Enter the Name as **WAIT_INTERVAL** and set it to value 1.
10. Click **Tools > Rule Manager** on the tool bar. Add a new rule, and name it Loop Counter. In the Expression box, type:
LoopCount < 10
11. Click **OK** twice to save your work and exit.

(Continued on next page)

Exercise 7–9 Wait Service

(Continued)

Instructions (Continued)

12. Click the edge line between **Choice Start** and the **LoopStart Sequence Start**.
13. Click **Add** in the Edge Editor. Select your rule from the drop-down, and choose **True** for the value.
14. Click the **Assign** that follows the LoopStart Sequence Start, and set the values:
From = LoopCount + 1
To = LoopCount
15. Click the **Repeat** activity, and set the values:
Name = CountingLoops
Ref = Loop
16. Click the last **Assign** in the top branch of the All Start and set the values:
Constant = Loop is done
To = LoopStatus
17. Click the last **Assign** in the business process and set the values:
Constant = Both branches have executed
To = FinalContext
18. Save and validate the business process with the name **Looping_Test**.
19. Check in the process with the name Looping_Test.
20. Execute your business process (no input file is needed).
21. Analyze the results.

(Continued on next page)

Lesson Review

Completed Objectives

This lesson was designed to help you to:

- List and explain the four basic service types
 - Explain how to create a service configuration
 - Explain Bootstrapping
 - Describe how the file system adapter works
 - Describe how OnFault Processing works
 - Explain how the Command Line adapter functions
 - Describe and demonstrate the use of a Lightweight JDBC adapter
 - Describe how the Invoke Business Process service works
-

Case Study

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Background

Introduction

This lesson provides an opportunity to analyze a business problem and create a business process based on the problem.

Lesson objectives

This lesson is designed to help you to:

- Analyze a business problem
 - Create a business process to solve the business problem
-

The Business Problem

Introduction

The Company agrees to perform a Vendor Managed Inventory service for two customers, My Little Store (MLS) and The Other Guy (TOG), both retailers are merchandise for the company.

The company auditor, requested to maintain a copy of the original data, so we must move a copy of the file to an archive directory. All processing should be automated without manual handling of data. It is already communicated and the data file(s) are now placed in /home/student/fscoll.

This exercise helps you understand more about bootstrapping, creating rules, XPath statements, and configuring different Services.

There are a number of different solutions that you can use to solve this case study, but one is given in detailed steps in this lesson.

The example solution that is presented in the detailed steps of this case study will have you build bootstrapping on a FSA into a business process instead of scheduling a FSA configuration like we did in the previous lesson. Both solutions work but scheduling is more commonly used in the real world. This solution does show the flexibility that is available with business processes and services.

Important!	When using bootstrapping or invoke, the business process that is invoked should be built, saved, and checked-in before it is referenced or used. This affects the order that Processes may be built, as you see in the following exercise.
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Important!	These exercises will have you make changes to service configurations in the GPM. Make sure to check Override Service Configuration under Options -> Preferences -> Service Editor. Please note that when overriding service configurations the overrides will only affect the bp where they are made. They will not globally affect other bp's that use that configuration.
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The Business Problem

(Continued)

Introduction (Continued)

In this solution, the following four different Business Processes need to be created:

1. **The Main Business Process:** Here build a business process that determines the Customer Id and take one of two paths, path for “My Little Store” or path for “The Other Guy”. This process invokes another Business Process that will determine whether it requires to order more items. After the business process runs, write out the file into the directory **/home/student/fs_output**. Suggested name: Main.BP.
 2. **The Invoke Business Process Service:** Here build two different Customer business processes to determine whether it needs to order more items. One business process for the “My Little Store” and another for “The Other Guy”. The order level for My Little Store is less than 10, then it should send more items. The order level for The Other Guy less than 20. Suggested names: LittleStore.BP and OtherGuy.BP
 3. **The Command Line Adapter Process:** Here build a business process that copies data into the correct folders, by calling the script, copytest.bat. This script looks for files in /home/student/fscoll with a name of *.xml. Then, move one copy of the files to /home/student/fsprocess and one copy to /home/student/fsarchive.
 4. **The File System Adapter Process:** Here build a process to PICK UP the files from /home/student/fsprocess and cause the Main.BP process to start running by using bootstrapping.
-

Exercise 8–1 Your Assignment

Instructions	For ease of understanding, this procedure is covered in three parts.
---------------------	--

- Setup Steps
- The Main Business Process
- Procedure for the Main Process

Setup Steps	■ Verify whether the following directories were created: ■ /home/student/fsarchive ■ /home/student/fsprocess ■ /home/student/fsoutput ■ Verify that script copytest.sh is in the directory /home/student.
--------------------	---

The Main Business Process	This process determines the Customer ID and then takes a path for “My Little Store” or “The Other Guy”. Then invoke a Business Process Service that determines whether it is required to order more items. At the end of the process, write file(s) into the directory /home/student/fs_output . Suggestion: Build LittleStore.BP and OtherGuy.BP first (go to Invoke Business Process Service / Per Customer), then come back and build this process.
----------------------------------	---

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Exercise 8–1 Your Assignment

(Continued)

The Main Business Process (Continued)

In this process, the following stencils are required.:

Start (1) Choice Start (1)

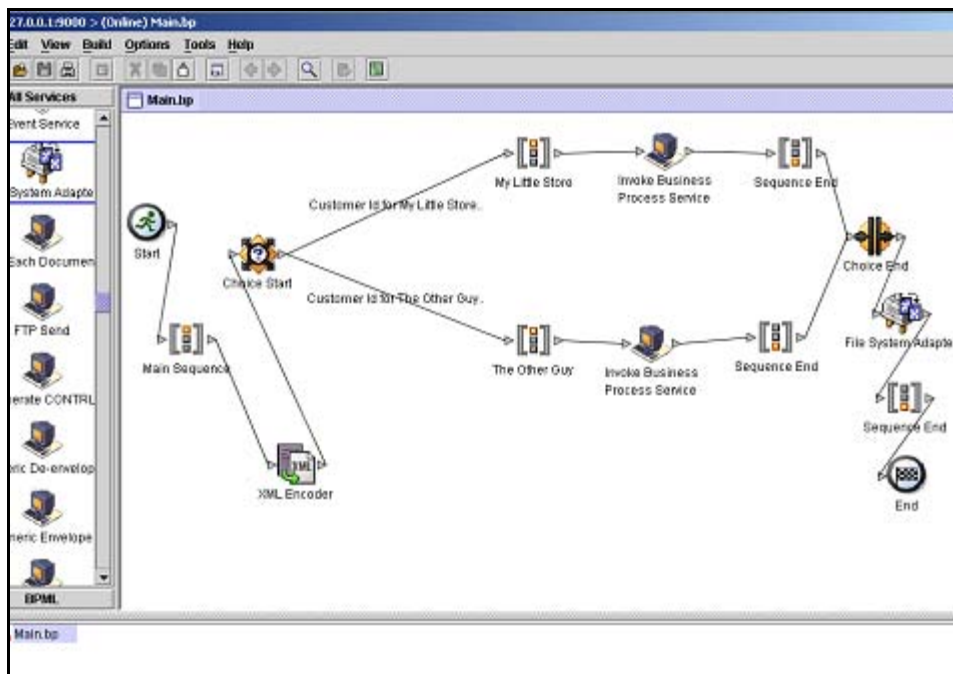
End (1) Choice End (1)

Sequence Start (3) XML Encoder Service (1)

Sequence End (3) File System Adapter (1)

Invoke Business Process (2)

The following figure depicts an example process.



(Continued on next page)

Exercise 8–1 Your Assignment

(Continued)

Procedure for the Main Process

You can follow the mentioned procedure for the main process.

1. Drag and connect your Stencils.
2. Changing the name of the 3 sequence starts.
 - a. Double-click the first sequence start and type in “Main Sequence” under the value column.
 - b. Double-click the top sequence start and type in “My Little Store” under the value column.
 - c. Double-click the bottom sequence start and type in “The Other Guy” under the value column.
3. Configure the XML Encoder:
 - a. Mode = Use existing XML document.
 - b. Output to process data = Yes.
4. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check for MLS**”. In the expression box, type the following Xpath string:

```
//INVENTORY/CUST_ID/text()='My Little Store'
```

(Continued on next page)

Exercise 8–1 Your Assignment

(Continued)

Procedure for the Main Process (Continued)

5. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check for TOG**”. In the expression box, type the following Xpath string:

```
//INVENTORY/CUST_ID/text()='The Other Guy'
```
6. Double click the line between **Choice Start** and the top **Sequence Start**. Click **Add** to open the name and value columns in the top Edge Editor. Select “**Check for MLS**” for the name and **true** under the value column.
7. The Invoke Business Process Service.
 - a. Invoke_mode = Sync
 - b. WFD_Name = LittleStore_#.BP or (whatever you named the business from the below exercise).
8. Double click the line between the **Choice Start** and the bottom **Sequence Start**. Click **Add** to open the name and value columns in the bottom Edge Editor. Select “**Check for TOG**” for the name and **true** under the value column.
9. The Invoke Business Process Service.
 - a. Invoke_mode = Sync
 - b. WFD_Name = OtherGuy.BP or (whatever you named the business from the below exercise).
10. The File System Adapter Service.
 - a. Action = Extraction.
 - b. AssignFilename = Use the original filename
 - c. Bootstrap = No
 - d. ExtractionFolder = /home/student/fsoutput
11. Save as Main.BP, validate and check in the business process.

Congratulations you can now add business process building to your Resume, but you are not done yet. Continue with the File System Adapter Process.

(Continued on next page)

Invoke Business Process Service Per Customer

Overview These two Customer business processes, My Little Store (MLS) and The Other Guy (TOG), will be invoked by the Main business process. It is required to build two copies of business process, one for “My Little Store” and another for “The Other Guy”. Build one, save, and validate; then do a Save As to create the second one with the other name.

In this process, the following stencils are required:

Start (1) Choice Start (1)

End (1) Choice End (1)

Sequence Start (3) Assign (2)

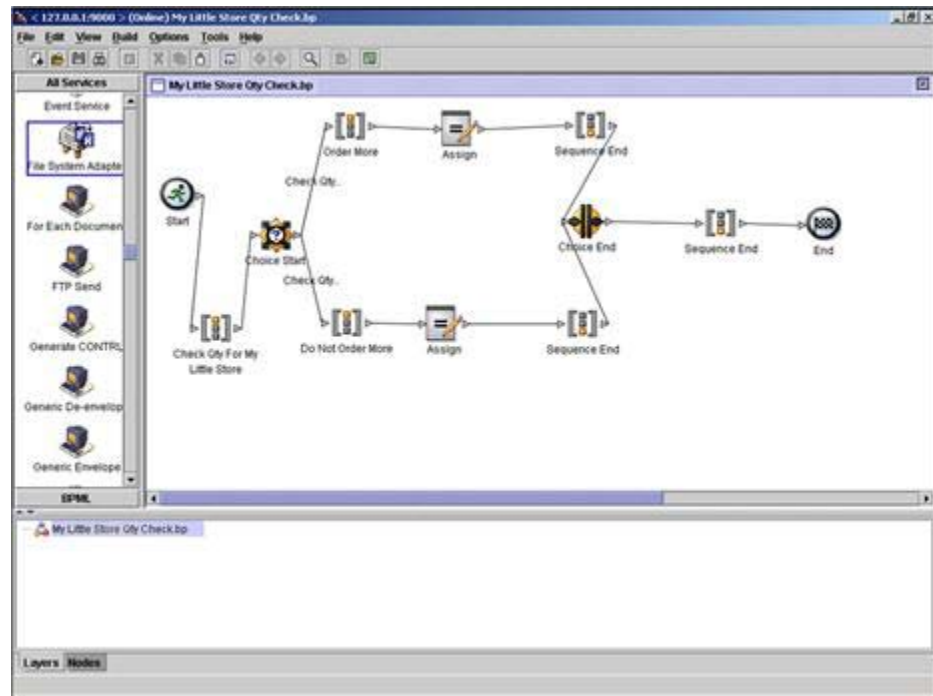
Sequence End (3)

(Continued on next page)

Invoke Business Process Service Per Customer (Continued)

Overview (Continued)

The following figure is an example of the process:



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Invoke Business Process Service Per Customer (Continued)

Step 1 for Customer Process

You can invoke business process per customer by following the steps that are listed:

1. Drag and connect your Stencils.
2. Changing the name of the 3 sequence starts.
 - a. Double-click the first sequence start and type in “Check QTY for My Little Store” in the value column.
 - b. Double-click the top sequence start and type in “Order More” under the value column.
 - c. Double-click the bottom sequence start and type in “Do Not Order More” under the value column.
3. Click **Tools > Rules Manager**. Click **Add** to bring up the Rule Editor. Populate the name box with “**Check QTY**”. In the expression box, type the following Xpath string:

```
sum(INVENTORY/PRODUCT/QTY) > 10
```
4. Click **Add** to open the name and value columns in the top Edge Editor. Select “**Check QTY**” for the name and **not true** under the value column.

(Continued on next page)

Invoke Business Process Service Per Customer (Continued)

Step 1 for Customer Process (Continued)

5. Click the **Assign** activity in Order More sequence and following information.

Name	Value
Constant	Order More Items
To	Results

6. Click **Add** to open the name and value columns in the bottom Edge Editor. Select "**Check QTY**" for the name and **true** under the value column.
7. Click the **Assign** activity in Do Not Order More sequence and following information.

This creates TAG in the output file: <results> Do not order more items</results>.

Name	Value
Constant	Do Not Order More Items
To	Results

8. Save as LittleStore.BP, validate and check in the business process.

(Continued on next page)

Invoke Business Process Service Per Customer (Continued)

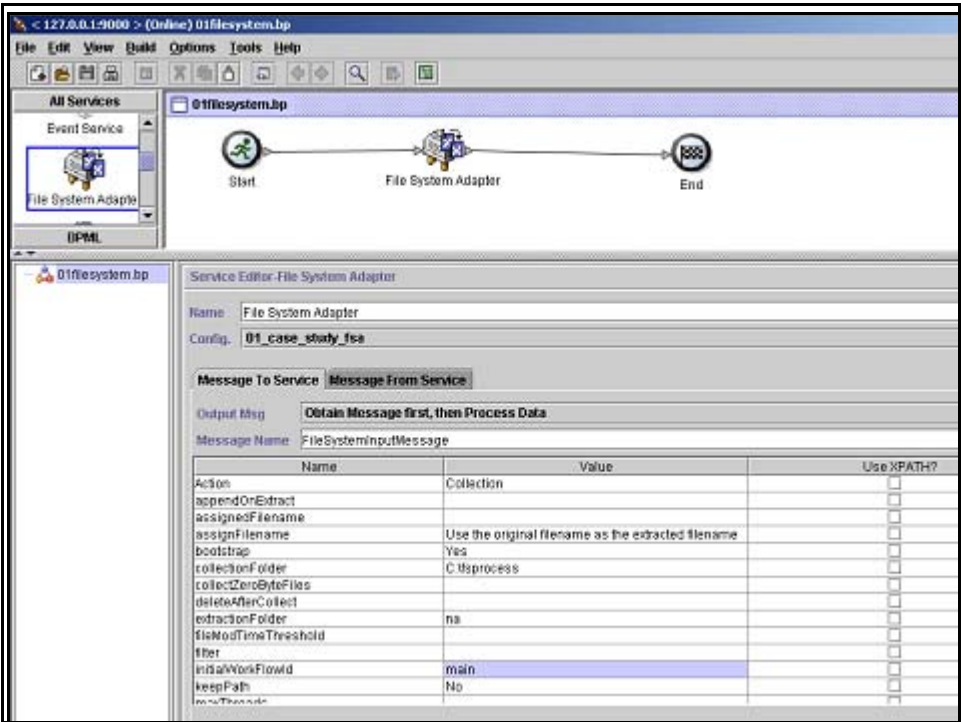
Step 1 for Customer Process (Continued)

Perform a **Save As** to create a second business process for “The Other Guy”. Remember that the Check QTY is different for this customer, (20) and the starting sequence should be named “Check QTY for The Other Guy”.

1. Save as **OtherGuy.BP**, validate, and check in the business process.
 2. Now the Main Process.
-

The File System Adapter Process

- Overview
- This process picks up the files from /home/student/fsprocess and then starts the Main business process. In this process, the following stencils are required:
Start (1)
End (1)
File System Adapter (1)
The following figure depicts an example process.



(Continued on next page)

The File System Adapter Process

(Continued)

Overview (Continued)

3. Drag and connect your Stencils.
 4. Open the Service Editor for the File System Adapter Service.
 - a. Action = Collection
 - b. Assign Filename = Use the original filename.
 - c. Bootstrap = Yes
 - d. CollectionFolder = /home/student/fsprocess
 - e. DeleteAfterCollect = Yes
 - f. InitialWorkFlowid = main.bp
 - g. FileModTimeThreshold = 0
 5. Save as **KICKOFF.BP**, validate and check in the business process.
-

The Command Line 2 Adapter Process

Overview

This process starts the business flow by calling a script that is called, copytest.sh. This script is set to look for files in /home/student/home/student//fscoll with a name of inventoryof3a.xml. After picking up the files, the script will make two copies of the file, depositing one file in /home/student/fsachive and the other in /home/student/fsprocess and then starting the File System Adaptor business process. (the previous process)

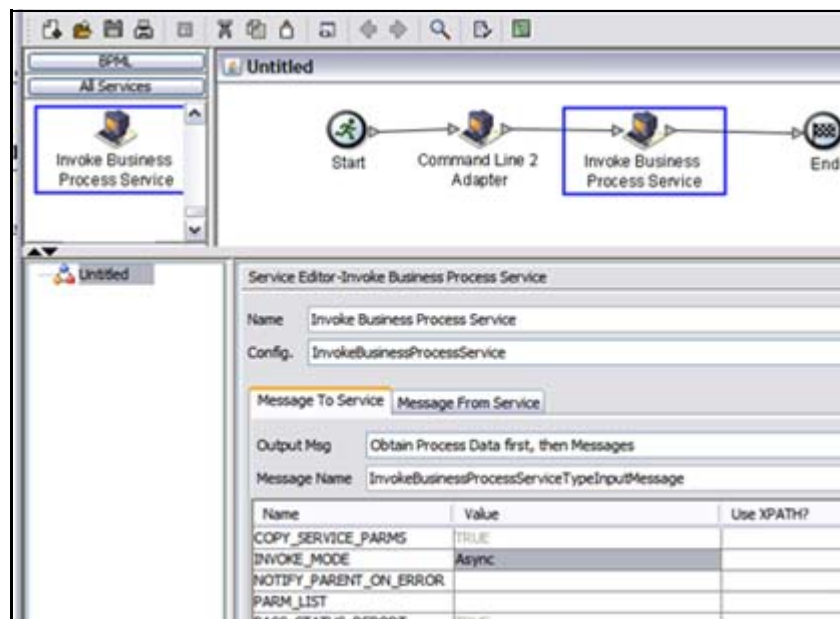
In this process, the following stencils are required:

Start (1)

End (1)

Command Line 2 Adapter (1)

The following figure depicts an example process.



(Continued on next page)

The Command Line 2 Adapter Process (Continued)

Overview (Continued)

1. Drag and connect your Stencils.
2. Open the Service Editor for the Command Line 2 Adapter Service.
 - a. Bootstrap = No
 - b. CmdLine = /home/student/copytest.sh
 - c. WorkingDirectory = /home/student.
 - d. InputDelete = true
 - e. OutputDelete = No
 - f. WaitOnProcess = Yes
3. Add an Invoke Business Process stencil between the command line 2 Adapter and the end stencil. Set the following parameters.
 - a. WFD_NAME = KICKOFF.bp (the bp from the previous step)
 - b. INVOKE_MODE = Asynchronous
4. Save as START.bp, validate, and check in the business process.

You can modify existing Service Configurations, or create new Service Configurations for use in this case study.
5. Locate the data files in the Case Study Labfiles folder under the 915 Fundamentals Folder and copy/paste all of the MLS and TOG files into the collection folder (/home/student/fscoll).

When you cause START to run manually, copytest.sh copies the XML files to the fsarchive directory and the fsprocess directory; then it uses the Invoke Business Process to start Kickoff. Kickoff pulls in the files from fsprocess and starts the Main Business Process.

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Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Analyze a business problem
 - Create a business process to solve the business problem
-